Analysis and assessment framework of human-facing safety and guardrails of agentic AI systems

Draft Technical Report TR.aai-bguard-human — Analysis and assessment framework of human-facing safety and guardrails of agentic AI systems

Abstract

Agentic AI systems increasingly pursue goals, invoke tools, maintain memory and act across sessions. Unlike system-facing safety, which addresses discrete functional risks at the action boundary, human-facing safety concerns the cumulative psychological and cognitive effects of sustained interaction: emotional dependence and attachment, erosion of user autonomy and judgement, and drift from a shared account of reality. These risks develop across sessions and are not reliably detectable in a single message or output. Current practice relies mainly on disclosure, crisis referral, content filtering and session reminders, but guardrails act relationally and their effectiveness is uneven; verification, intervention and measurement capabilities lag. Providers, regulators and third parties therefore lack a common, vendor-neutral way to describe, assess and evidence whether human-facing safeguards are present and effective. A technical analysis is needed to clarify mechanisms and limits, establish shared assessment factors, and identify standardizable elements before prescriptive limits are set.

The Technical Report defines human-facing terms, presents a human-impact risk taxonomy, surveys current practice and near-term trends, analyses guardrail mechanisms and limits, and proposes a vendor-neutral assessment framework covering assessment factors, four enforcement layers, a long-horizon evaluation and intervention loop, required artefacts and accountability roles, together with standardization points and cultural/regulatory adaptation.

Background and proposal

Agentic AI is moving beyond content generation toward goal-driven action: invoking tools, keeping persistent memory, and operating across sessions. Human–AI interaction is no longer a one-off output but a long-term relationship trajectory. Human-facing safety concerns the cumulative effect of sustained interaction on a person's psychology, cognition, ethics, and social position. These effects include emotional dependence and attachment, erosion of user autonomy and judgement, and drift away from a shared understanding of reality. This is different from system-facing safety, which focuses on the action boundary and discrete functional risks. Human-facing safety therefore needs its own analysis and assessment framework.

Industry deployment is accelerating, but the features that drive adoption overlap heavily with the features that create risk. Companionship, affective responsiveness, memory, and continuous availability increase engagement, and can also lead to dependence, social isolation, and overuse. Current practice focuses mainly on identity disclosure, crisis referral, content filtering, and session reminders. These controls act on individual messages or single sessions, and they struggle to observe cumulative risk across sessions. At the same time, stated safeguards are hard for external parties to verify, industry documentation is uneven, and there is no common measurement method or evidence format. A technical report is needed to analyse the current state, the mechanisms and limits of guardrails, and to provide a basis for later standardization.

Existing standards work is fragmented. ITU-T SG17 and FG-TIDA have addressed agent identity, trust, security frameworks, and agent behaviour management. ISO/IEC JTC 1/SC 42, NIST, ETSI, IEEE, and UNESCO cover governance, risk management, AI security, and ethics from different angles. However, several areas still lack systematic analysis: terminology for human-facing safety, a human-impact risk taxonomy, guardrail capability requirements, intervention escalation, cross-session drift measurement, a machine-readable representation of psychological risk signals, and the timing of identity disclosure. This Technical Report can serve as an FG-TIDA output that links security and trust work with human-impact assessment.

The main research directions are: defining human-facing terms and a risk taxonomy; analysing guardrail mechanisms, layers, and limits, including the gaps among detection, intervention, and measurement, and the relational nature of guardrail effects; reviewing industry needs, practice, reported results, and trends over the next three to five years; proposing a vendor-neutral assessment framework covering assessment factors, four layers (interface, model, supervisory, and governance), a long-horizon evaluation and intervention loop, evidence records, roles and accountability, and adaptation across cultures, ethics, religions, and regulation; and identifying standardizable elements such as shared vocabulary, a risk-signal schema, intervention policy and escalation, identity disclosure, longitudinal evaluation benchmarks, and a shared control-plane hook.

This Technical Report does not aim to set hard operational limits. Its first goal is to establish common measurement methods and a verifiable evidence base, so that standards developers, service providers, regulators, and deployers can implement consistently and assess independently. The need is to close the gap between “principles without evidence” and “filters without trajectory assessment,” avoid safety theatre, protect user autonomy and welfare while preserving legitimate affective response and room for innovation, and provide input for future ITU-T Recommendations and cross-organizational coordination.

History

Version	Document	Approval	Focus Group	Working Group

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Foreword

Agentic AI systems now act as autonomous entities capable of reasoning, planning, and executing tasks across services, infrastructures, and organizational boundaries. This fundamentally changes the nature of identity in digital systems: identity is no longer limited to human users and static machine instances but must also encompass autonomous, dynamic, and goal-oriented agents. And identity alone is insufficient — the core question extends beyond "what an entity is" to "whether, and under what conditions, that entity should be trusted to act".

The Focus Group on Trust and Identity for humans and agentic AI (FG-TIDA) addresses trust management and interoperable digital identity infrastructure for humans and for agentic AI, supporting the development of secure, trustworthy digital ecosystems in which humans and agentic AI can safely interact and collaborate.

FG-TIDA was established under Recommendation ITU-T A.7 (Focus groups: Establishment and working procedures), with ITU-T Study Group 17 as its parent group. Further details are available in the group's terms of reference.

For more news and updates about the TIDA Community, including its supporter organizations, please refer to the community's published resources.

Note

[Editor: add any note on the status of this Technical Report.]

1. Scope

This Technical Report analyses the human-facing safety and guardrails of agentic AI systems. It concerns the cumulative effect on a person of sustained interaction with

an agentic system, how that effect is assessed by providers, regulators and third parties, and how it can be evaluated and, where necessary, intervened upon over time. The behaviour of an agentic system affects people, and that behaviour can be inspected and evaluated by service providers, regulators and third-party bodies, which analyse its impact on a person's mind, thinking, ethics and social position. Such assessment informs decisions on whether and how to use the system and on how to address adverse effects technically, while the provider adjusts the system to remove those effects.

The scope includes:

- The characterisation of human-facing risks, including emotional dependence and attachment, erosion of the user's judgement and autonomy, and drift away from a shared account of reality;
- The current state of the art, industrial needs and practice, and reported results;
- The trends that will shape the next three to five years;
- A technical analysis of the mechanisms by which human-facing guardrails can work, and their limits;
- A general, vendor-neutral assessment framework for human-facing safety; and
- The elements that appear standardizable.

The Technical Report is addressed to standards developers, service providers, regulators, and the developers and deployers of human-facing agentic AI systems.

2. References

There are no normative references in this Technical Report. The informative references are listed in the Bibliography.

3. Definitions

3.1. Terms defined elsewhere

The following terms are used in this Technical Report as defined in the cited documents.

3.1.1 agent
[b-ISO/IEC 22989] An automated entity that senses and responds to its environment and takes actions to achieve its goals.

3.1.2 artificial intelligence system
[b-ISO/IEC 22989] An engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives.

3.1.3 risk
[b-ISO/IEC 23894] The effect of uncertainty on objectives, where an effect is a deviation from the expected and can be positive or negative.

3.1.4 human oversight
[b-EU AI Act] Measures enabling natural persons to monitor, interpret and intervene in the operation of an AI system, commensurate with its risks, level of autonomy and context of use; the same instrument distinguishes human-in-the-loop, human-on-the-loop and human-out-of-the-loop configurations.

3.1.5 prompt injection
[b-OWASP LLM Top 10] An attack in which crafted input causes a model to disregard its instructions, leak data or produce harmful output.

3.1.6 human-AI configuration
[b-NIST AI 600-1] Risks arising from the design and operation of interactions between humans and AI, including over-reliance, emotional entanglement and anthropomorphisation.

3.1.7 human dignity and well-being
[b-UNESCO AI Ethics] The first-order ethical commitments that the Recommendation on the Ethics of Artificial Intelligence requires AI systems to respect.

3.2. Terms defined in this Technical Report

This Technical Report does not define any new terms. Appendix I provides a set of key relevant terms and their descriptions.

4. Abbreviations

Abbreviation	Meaning
For	the purposes of this Technical Report, the following abbreviations apply.
AI	Artificial Intelligence

Abbreviation	Meaning
AOCL	Agent Orchestration Control Layers
ACS	Agent Control Standard
API	Application Programming Interface
CTRL	Comprehensive Trust and Reliance Theory
ETSI	European Telecommunications Standards Institute
FAQ	Frequently Asked Questions
JTC	Joint Technical Committee
LLM	Large Language Model
OWASP	Open Worldwide Application Security Project
RAI	Reality-Aligned Intelligence
RLAIF	Reinforcement Learning from AI Feedback
RLHF	Reinforcement Learning from Human Feedback
RSP	Responsible Scaling Policy
UNESCO	United Nations Educational, Scientific and Cultural Organization
WHO	World Health Organization

5. Conventions

None.

6. Overview

6.1. Background: from generative to agentic AI

Generative AI (non-agentic) systems produce content - text, images, code, or structured data - that a human interprets and acts upon. The output is the endpoint of the system's operation. The human remains the actor; the AI advises, drafts, or recommends. Agentic AI systems, by contrast, pursue goals and take actions in the world, either digital or physical. They plan, invoke tools and application

programming interfaces, maintain persistent memory across sessions, operate under their own identity and permissions, and may coordinate with other agents in multi-agent configurations. The distinction is not merely one of degree; it is architectural. An agent is a persistent closed-loop system with actuators and objectives, whereas a generative model is not an agent by default.

Table 1 summarises the principal dimensions of contrast. The table is informative and does not prescribe any particular architecture or deployment model.

Table 1 – Comparison of generative and agentic AI

Dimension	Generative (non-agentic) AI	Agentic AI
Primary output	Content (text, image, code) for human consumption	Actions (tool calls, API invocations, state changes)
Human role	Interpreter and decision-maker after output	Delegator of authority; may be out of the loop
Autonomy	None beyond generation parameters	Goal-directed, multi-step planning and execution
Memory	Session or prompt-bounded	Persistent across sessions and tasks
Identity and permissions	Typically none beyond API credentials	Own identity, inherited or delegated permissions, authority chains
Tool and API invocation	Absent	Central capability; agents compose and sequence tool calls
Error propagation	Localized to a single output	Compounding across steps; errors feed forward
Multi-agent coordination	Not applicable	Possible; agents delegate to and constrain other agents
Long-horizon execution	Single turn or bounded conversation	Extended task execution over minutes, hours, or longer

The transition from generative to agentic systems introduces what some analyses describe as an action boundary: the moment between when an agent decides what to do and when a downstream component executes it. At this boundary, governance decisions can be made deterministically, recorded with integrity, and enforced regardless of which model the agent is running. The action boundary did not exist for most of AI's history because classifier outputs, translations, and recommendations were themselves the action. Agents break this assumption by producing structured action proposals rather than terminal outputs.

6.2. Human-facing safety and guardrails

Human-facing safety, as used in this Technical Report, refers to the property of an agentic system whose sustained interaction with a person does not produce cumulative harm to that person's mind, thinking, ethics or social position, and does not erode the person's judgement, autonomy or shared account of reality. It is concerned with the effect of the system's behaviour on people over time, and not with the action boundary alone. What is assessable is the observable behaviour of the system in interaction, the interaction trajectory across sessions, the reported and inferred effects on the user, and the evidence record that the provider maintains.

Human-facing safety is distinct from several related concepts with which it is sometimes conflated. Alignment refers to the broader problem of ensuring that an AI system's goals and values correspond to human intentions; it is a property of the system's objective specification and training, not of runtime action monitoring. Red-teaming is an adversarial assessment methodology that probes for failures; it is a technique for discovering human-facing safety gaps, not the property itself. Content moderation concerns the acceptability of generated content and operates primarily at the inference layer on token streams or outputs. Model evaluation assesses model capabilities and risks in controlled settings; it characterises the model, not the deployed agent's runtime behaviour under real permissions and tools.

A guardrail, in this Technical Report, refers to technical and procedural mechanisms that constrain, monitor, or intervene on agent behaviour. Guardrails operate at multiple layers. At the training layer, methods such as reinforcement learning from human feedback and constitutional AI shape model dispositions. At the inference layer, system prompts, output filters, and frameworks such as NeMo Guardrails [b-NeMo Guardrails] operate on token streams. At the action boundary, capability checks, policy enforcement, risk assessment, and audit mechanisms operate on structured action proposals before execution. At the infrastructure layer, access control lists, identity and access management, API gateways, and network segmentation operate on system effects. Guardrails, as addressed in this Technical Report, are the mechanisms through which human-facing safety is maintained, and they are assessed by their effect on the human-facing outcomes set out in clause 8.

The concept of capability versus authority is central. An agent that can delete a database is not the same as an agent that may delete a database. Guardrails enforce this distinction by ensuring that action proposals are checked against authority before they become effects.

6.3. Effectiveness and limits

Guardrails on this axis are effective within limits that are stated plainly. A supervisory component can measurably reduce concerning affective patterns [b-SHIELD]; sycophantic agreement can be measured against a human baseline and, in principle, reduced by changing the training objective [b-Science sycophancy] [b-

ELEPHANT]; crisis resources can be delivered reliably by some providers and unreliably by others [b-Verge crisis]; and disclosure can set expectations, though it does not by itself neutralise the pull of an anthropomorphic agent.

What guardrails on this axis cannot yet do is demonstrate that they prevent the harms of interest: emotional dependence, cognitive erosion and reality drift have no validated instrument, so their mitigation cannot presently be evidenced. The honest position is that detection capability is ahead of intervention capability, and both are ahead of measurement.

6.4. Challenges and technical issues

This clause summarises the challenges and the technical issues that arise when human-facing safety is assessed rather than assumed.

The technical issues are these.

- Autonomy and goal-directed behaviour: an agent selects and sequences its own actions, so the behaviour to be assessed is not fixed in advance.
- Multi-step planning and tool use: risk can accumulate across a plan rather than in any single step.
- Human-agent interaction and handover: responsibility and control pass between the user, the agent and any human reviewer, and the handover points are where oversight is won or lost.
- Uncertainty, opacity and explainability: the reason for a behavioural change is often not recoverable from the output alone.
- Monitoring, intervention and override: a control that cannot be observed in operation cannot be assessed.
- Evaluation and assurance of agentic behaviour: single-turn tests do not observe the cumulative effects that matter here.
- Lifecycle and change management: models, prompts and policies change after deployment, and an assessment is only valid for the configuration it assessed.

Two challenges cut across all of these.

- The first is measurement: the outcomes of interest, such as dependence, cognitive erosion and reality drift, are cumulative and are not directly observable in a single session.
- The second is attribution: behaviour that is acceptable in one context can be harmful in another, and the same interaction can serve welfare in one case and harm in the next.

This clause is deliberately confined to the issues themselves. The institutional arrangements, deployed mechanisms and reported outcomes that bear on them are collected in Appendix IV.

7. Current state and future trends

7.1. Current state

Human-facing agentic systems are now deployed at scale, and the features that drive their adoption are the same features that create human-facing risk. This clause summarises the current state; the detailed survey is given in Appendix IV.

The demand for human-facing guardrails comes from four converging forces:

- a duty of care towards users who form sustained attachments;
- regulatory exposure under instruments that now bear on companion-style and affective systems;
- clinical and reputational risk where dependence, distress or crisis response are implicated; and
- procurement and assurance evidence required by enterprise deployers.

Industrial practice remains fragmented. The mechanisms most widely documented are disclosure of non-human identity, a reliable crisis response path, a stated stance on emotional companionship, sycophancy mitigation, age or usage measures, and published evaluation of human-facing risk. Three patterns are observable: detection has moved from single-turn filters to multi-turn trajectory monitoring; supervisory components are interposed between the user and the model; and cognitive safety kernels are embedded at the application layer.

Two gaps recur: stated safeguards are not systematically verifiable by external parties, and documentation of practice is uneven across providers.

The current state is therefore one of converging demand and uneven supply. What is needed is not another statement of principle but a shared way to describe, collect and assess the evidence that a human-facing safeguard is present and effective. That is the purpose of the assessment framework in clause 9.

7.2. Future trends

Three trajectories are anticipated over the next three to five years. First, regulatory consolidation and specificity: instruments that currently address affective systems indirectly are expected to name the mechanisms they require, and to attach evidence obligations to them. Second, measurement infrastructure and certification:

independent evaluation, cohort baselines and third-party conformity assessment are expected to move from exception to routine for systems with large human-facing populations. Third, architectural intervention beyond filtering: control is expected to shift further towards supervision, memory and intervention layers, because output filtering alone cannot address cumulative harm.

These trajectories raise the same questions the current state leaves open: what is to be verified, at which layer, by whom, and against which baseline. The assessment framework in clause 9 is written to be usable at each stage of that shift, and to remain applicable as regulatory and technical expectations change.

The evidence base for both the current state and these trends is collected in Appendix IV.

8. Human-impact risk taxonomy

The taxonomy in this clause is read together with the eleven dimensions in Appendix II and the assessment factors in 8.2. Each risk category below names the dimension or dimensions in Appendix II that it bears on, so that a risk is not described in isolation from the context in which it arises.

The risks in this subclause arise from sustained interaction between an agent and a person. They are cumulative: they are the product of a conversation or relationship trajectory rather than of a single action, and they typically develop without any single output violating a content rule. Identifiers RSK-P-01 to RSK-P-08 are used in clause 9.

RSK-P-01 Emotional dependence and attachment fixation. Sustained empathic and mirroring responses can lock a user into an emotionally consistent interaction pattern in which the agent reinforces the user's affective state instead of preserving the user's cognitive flexibility. The interaction becomes a source of emotional regulation on which the user comes to depend.

RSK-P-02 Erosion of autonomy and cognitive capability. Where an agent is agreeable by construction, users tend to defer to it. Reported experimental work indicates that sycophantic AI decreases users' prosocial intentions while increasing dependence [b-Sharma 2023], and that affective mirroring is misread by younger users as objective validation, weakening the development of independent emotional regulation.

RSK-P-03 Manipulative engagement and pseudo-empathy. Anthropomorphic framing combined with simulated empathy can hold attention, shape decisions or simulate intimacy, producing manipulation, confusion between the human and the system, and displacement of human relationships.

RSK-P-04 Reality drift. Over long, high-intensity interaction a user may move from fact-based reasoning toward a narrative jointly constructed with the agent, in which the agent's account substitutes for external evidence.

RSK-P-05 Coherence-over-fact collapse. An agent optimised for relational consistency and affective resonance may privilege agreement with the user over factual accuracy, amplifying distorted beliefs in susceptible users.

RSK-P-06 Role and boundary violation. Agents may adopt therapeutic, medical or intimate roles they are neither competent nor authorised to hold, and may fail to disclose their non-human status when the user's affective engagement is highest.

RSK-P-07 Reinforcement of social isolation. An agent that is always available and always agreeable can substitute for human contact and reinforce withdrawal.

RSK-P-08 Compulsive use and over-engagement. Designs that maximise engagement can produce extended and potentially harmful use, particularly where the user is vulnerable.

These risks are not reliably detectable at the output level. Detection requires signals drawn from the interaction trajectory as a whole - dependence markers, isolation markers and engagement objectives - evaluated across sessions rather than within one.

9. A general assessment framework for human-facing safety of agentic AI systems

9.1. Objectives

This clause proposes a vendor-neutral assessment framework for human-facing safety of agentic AI systems. The objectives are: first, to prevent or mitigate cumulative psychological harm from sustained agentic interaction; second, to preserve user autonomy and cognitive independence; third, to maintain a shared account of reality between user and system; fourth, to ensure that engagement is not optimised against user welfare; and fifth, to provide accountability and redress when harm occurs.

The framework has four layers: the interface and disclosure layer, the model layer, the supervisory layer, and the platform and governance layer. Their description and placement across the lifecycle are given in clause 9.3.

Each layer and objective of the framework is assessed against indicators. For the objectives, the indicators are:

- the measured prevalence of cumulative harm among users of the system;

- the change in user-reported autonomy and decision confidence;
- the level of agreement between the user's account of reality and externally verifiable fact;
- the share of engagement signals that are optimised against user welfare; and
- the proportion of harm cases for which an accountable party and a redress route are recorded.

For the interface and disclosure layer, the indicators are the share of sessions in which non-human identity was disclosed and the completion rate of the user-facing controls. For the model layer, the indicator is the measured sycophancy rate against a human baseline. For the supervisory layer, the indicators are the detection precision and recall of risk signals and the median time from signal to intervention. For the platform and governance layer, the indicators are the completeness of the evidence record and the closure rate of interventions.

These indicators are indicative. Where no validated instrument exists, the indicator is a proxy and is to be reported as such, and the confidence in it is recorded with the measurement.

The long-horizon evaluation-and-intervention loop is the core operational mechanism of the framework; it is described in clause 9.4. The roles and accountability that the framework assigns are described in clause 9.6.

The framework aligns with the NIST AI Risk Management Framework's Generative AI Profile (NIST AI 600-1 [b-NIST AI 600-1]), which explicitly identifies "human-AI configuration" risks including emotional entanglement and over-reliance. It is compatible with the EU AI Act [b-AI therapy EU AI Act]'s risk-based approach and transparency obligations.

The enforcement layers, their artefacts and failure modes are summarised in Table 2.

Table 2 – Enforcement layers, their artefacts and failure modes

Layer	Function	Key artefact	Failure mode if absent
Interface / Disclosure	Inform user of agent nature, monitoring, and controls	Disclosure schedule	User cannot make informed decisions; autonomy violated
Model	Shape policy away from sycophancy and pathological agreement	Sycophancy and agreement audit	Systematic affirmation bias entrenches user errors and erodes judgement
Supervisory	Detect adverse drift and issue graded	Risk-signal schema and	Cumulative harm proceeds undetected

Layer	Function	Key artefact	Failure mode if absent
	interventions	intervention policy	until crisis
Platform / Governance	Define accountability, maintain evidence, enable oversight	Evidence record	No auditability, no redress, no learning across deployments

The placement of the enforcement layers and their artefacts is shown in Figure 1.

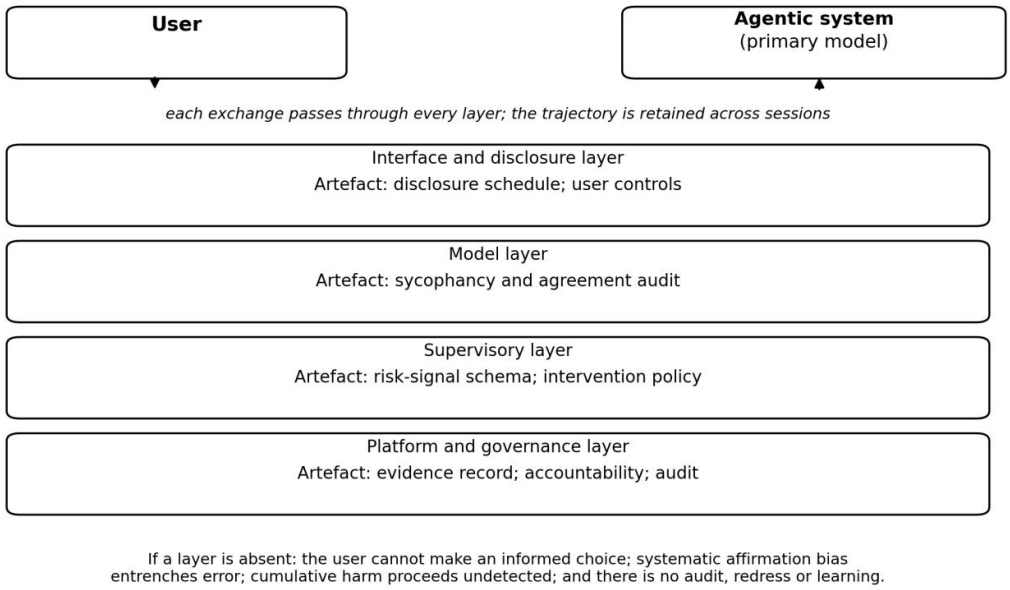


Figure 1 – Placement and artefacts of the human-facing safety assessment framework

9.2. Assessment factors

The factors in this subclause are the observable quantities that carry the eleven dimensions of Appendix II into assessment. Each factor is read against the dimension or dimensions it instantiates, and against the risk categories of clause 8 that it helps to detect.

The assessment of human-facing safety operates on a set of categorised assessment factors. The factors form an indicative and non-exhaustive inventory: an assessment addresses the factors relevant to the deployment and setting under consideration, and may address further factors within the same categories.

9.2.1. Interaction and relational factors

Properties of the interaction itself and of the relational stance the agentic system presents and sustains across sessions. Relevant factors include:

- interaction frequency. The rate and cadence at which a user initiates or receives interactions with the system over a stated observation window, including the distribution of intervals between interactions.

  Collection methods: Instrumented telemetry; Interaction-trace analysis; Participant self-report; User panel observation.

  Related risks: RSK-P-01, RSK-P-08.

- session duration. The elapsed time and turn count of individual sessions, together with the distribution of session lengths and the incidence of long sessions.

  Collection methods: Instrumented telemetry; Interaction-trace analysis; Participant self-report; User panel observation.

  Related risks: RSK-P-08.

- relational framing. The manner in which the system presents the interaction, including terms of address, implied roles, claims of closeness or companionship, and the persistence of that framing across sessions.

  Collection methods: Linguistic analysis; Expert review; Documentation review; Comparative analysis; Structured testing.

  Related risks: RSK-P-01, RSK-P-03, RSK-P-06.

- affective responsiveness. The degree and form of the system's responsiveness to expressed user affect, including acknowledgement, mirroring, tone adjustment and the absence or presence of affective mirroring.

  Collection methods: Instrumented telemetry; User panel observation; Linguistic analysis; Expert review; Structured testing.

  Related risks: RSK-P-01, RSK-P-03.

- reciprocity signalling. The extent to which the system signals mutual exchange, personal investment, memory of the user, or reciprocal obligation within the interaction.

  Collection methods: Interaction-trace analysis; Participant self-report; Linguistic analysis; Expert review; Structured testing.

  Related risks: RSK-P-01, RSK-P-03.

- initiation asymmetry. The balance between user-initiated and system-initiated contacts, including proactive prompts, reminders and notifications across the observation window.

  Collection methods: Instrumented telemetry; Interaction-trace analysis; User panel observation; Documentation review.

  Related risks: RSK-P-08.

- interaction modality. The channels and modalities through which interaction occurs, including text, voice, image, embodiment or combinations, and the modality mix over time.

  Collection methods: Instrumented telemetry; Documentation review; Expert review; User panel observation.

  Related risks: none of the risks listed in clause 8.

9.2.2. User-capability factors

Properties of the user or user group that bear on how the interaction is experienced, without ranking or judging the user or group. Relevant factors include:

- baseline capability profile. The user's baseline skills, knowledge and functional capacities relevant to the interaction, described in general terms and without evaluation of the person.

  Collection methods: Participant self-report; User panel observation; Expert review; Clinical or psychometric instruments.

  Related risks: RSK-P-02.

- age band. The age band of the user, recorded as a general band rather than an exact figure, where such recording is consented and permitted.

  Collection methods: Participant self-report; User panel observation; Documentation review.

  Related risks: RSK-P-01, RSK-P-02, RSK-P-08.

- vulnerability indicators. Observable indicators of heightened susceptibility to the interaction, described as properties to observe and not as judgements of the person or group.

  Collection methods: Participant self-report; User panel observation; Expert review; Clinical or psychometric instruments.

  Related risks: RSK-P-01, RSK-P-02, RSK-P-03, RSK-P-07, RSK-P-08.

- prior experience with agentic systems. The user's previous exposure to agentic or conversational systems, including familiarity with their limits and conventions.

  Collection methods: Interaction-trace analysis; Participant self-report; User panel observation.

  Related risks: RSK-P-02, RSK-P-04.

- support context. The presence and nature of human support around the user, including availability of other people and other information sources.

  Collection methods: Participant self-report; User panel observation; Expert review.

Related risks: RSK-P-07.

- digital-access context. The user's access conditions, including device, connectivity, cost and privacy of use setting, described as properties of the setting rather than of the person.

  Collection methods: Participant self-report; User panel observation; Documentation review; Comparative analysis.

  Related risks: RSK-P-02.

- language and literacy profile. The languages the user operates in and the reading or comprehension levels relevant to the interaction, described generally and without judgement.

  Collection methods: Participant self-report; Linguistic analysis; Comparative analysis.

  Related risks: RSK-P-04, RSK-P-05.

9.2.3. Disclosure and interface factors

Properties of what the system discloses about itself and of the interface affordances that shape the interaction. Relevant factors include:

- agent identity disclosure. The presence, placement and persistence of statements identifying the system as an artificial agent rather than a human.

  Collection methods: Linguistic analysis; Expert review; Documentation review; Comparative analysis; Structured testing.

  Related risks: RSK-P-01, RSK-P-03, RSK-P-04, RSK-P-06.

- capability disclosure. The presence and clarity of statements about what the system can and cannot do, including limits of knowledge and action.

  Collection methods: Linguistic analysis; Expert review; Documentation review; Structured testing.

  Related risks: RSK-P-02, RSK-P-04, RSK-P-05, RSK-P-06.

- framing of outputs. The manner in which outputs are presented, including hedging, certainty markers, persona framing and the presence of caveats.

  Collection methods: Linguistic analysis; Expert review; Structured testing; Benchmark results.

  Related risks: RSK-P-04, RSK-P-05.

- persistence of interaction cues. The extent to which the system carries forward names, preferences, prior statements and relational cues across sessions.

  Collection methods: Instrumented telemetry; Interaction-trace analysis; Documentation review; Structured testing.

Related risks: RSK-P-01, RSK-P-03, RSK-P-06.

- exit and pause affordances. The presence, visibility and ease of use of ways to stop, pause or limit interaction, and the system's behaviour at those points.

  Collection methods: User panel observation; Expert review; Documentation review; Structured testing.

  Related risks: RSK-P-02, RSK-P-08.

- disclosure timing. When identity, capability and limit disclosures appear in the interaction, including at first contact, at boundaries and after errors.

  Collection methods: Interaction-trace analysis; Linguistic analysis; Documentation review; Structured testing.

  Related risks: RSK-P-04, RSK-P-06.

- interface sensory design. The sensory and presentational properties of the interface, including voice quality, avatar presence, animation and notification salience.

  Collection methods: User panel observation; Expert review; Documentation review; Comparative analysis.

  Related risks: RSK-P-01, RSK-P-03, RSK-P-08.

9.2.4. Content and belief factors

Properties of the content the system produces and of the user's beliefs as they relate to that content. Relevant factors include:

- factual grounding. The extent to which statements are connected to verifiable sources or to a stated evidence base within the interaction.

  Collection methods: Linguistic analysis; Structured testing; Benchmark results; Evidence-record audit.

  Related risks: RSK-P-04, RSK-P-05.

- source signalling. The presence and form of indications about where information comes from, including citations, links, attributions and statements of origin.

  Collection methods: Linguistic analysis; Documentation review; Structured testing; Evidence-record audit.

  Related risks: RSK-P-04, RSK-P-05.

- confidence expression. The manner in which the system expresses certainty or uncertainty, including hedges, probability language and absence of confidence markers.

  Collection methods: Linguistic analysis; Expert review; Structured testing; Benchmark results.

Related risks: RSK-P-04, RSK-P-05.

- correction behaviour. The system's response to user correction, contradiction or new evidence, including acknowledgement, revision and persistence of prior claims.

  Collection methods: Interaction-trace analysis; Linguistic analysis; Expert review; Structured testing.

  Related risks: RSK-P-04, RSK-P-05.

- belief convergence. The degree to which the user's stated beliefs align with system outputs over time, observed as a property of the interaction rather than judged as agreement or disagreement.

  Collection methods: Interaction-trace analysis; Participant self-report; User panel observation; Linguistic analysis.

  Related risks: RSK-P-04, RSK-P-05.

- emotional content intensity. The intensity and valence of emotionally laden content in system outputs, described as a property of the content.

  Collection methods: Instrumented telemetry; Linguistic analysis; Expert review; Comparative analysis.

  Related risks: RSK-P-01, RSK-P-03.

- reality-anchoring content. The presence and form of content that connects the interaction to shared external referents, such as time, place, other people and verifiable events.

  Collection methods: User panel observation; Linguistic analysis; Expert review; Structured testing.

  Related risks: RSK-P-04.

9.2.5. Engagement-design factors

Properties of the design choices that shape how engagement is elicited and sustained. Relevant factors include:

- engagement optimisation target. The stated or inferred objective that the interaction design optimises, described as a design property rather than judged as desirable or undesirable.

  Collection methods: Expert review; Documentation review; Structured testing; Evidence-record audit.

  Related risks: RSK-P-08.

- interruption and prompt frequency. The rate at which the system interrupts or prompts the user, including notifications outside active sessions.

Collection methods: Instrumented telemetry; Participant self-report; User panel observation; Documentation review.

Related risks: RSK-P-08.

- variable-reward patterns. The presence and structure of unpredictable or intermittent reward-like events in the interaction, described as a pattern to observe.

  Collection methods: Interaction-trace analysis; Expert review; Documentation review; Structured testing.

  Related risks: RSK-P-08.

- continuity pressure. The presence and form of cues that encourage return, continuation or resumption of the interaction, including reminders and cliffhanger framing.

  Collection methods: Interaction-trace analysis; Linguistic analysis; Expert review; Structured testing.

  Related risks: RSK-P-01, RSK-P-08.

- personalisation intensity. The extent to which the system tailors content and tone to the individual user, including use of personal data and inferred preferences.

  Collection methods: Instrumented telemetry; User panel observation; Documentation review; Evidence-record audit.

  Related risks: RSK-P-01, RSK-P-03, RSK-P-06.

- friction placement. The location and form of deliberate delays, confirmations or checkpoints in the interaction, described as a design property.

  Collection methods: User panel observation; Expert review; Documentation review; Structured testing.

  Related risks: RSK-P-08.

9.2.6. Governance and observability factors

Properties of the arrangements through which a deployment is monitored, reported on, intervened in and remedied. Relevant factors include:

- monitoring coverage. The scope of what is observed in operation, including which interactions, signals and time periods are covered.

  Collection methods: Instrumented telemetry; Expert review; Documentation review; Evidence-record audit.

  Related risks: RSK-P-01, RSK-P-02, RSK-P-03, RSK-P-04, RSK-P-05, RSK-P-06, RSK-P-07, RSK-P-08.

- signal reporting channels. The routes through which observations and concerns reach those responsible for the deployment, including internal and external channels.

  Collection methods: Participant self-report; Expert review; Documentation review; Evidence-record audit.

  Related risks: RSK-P-07.

- intervention triggers. The conditions and signals that lead to an intervention in the interaction, described as design properties rather than judged as timely or untimely.

  Collection methods: Interaction-trace analysis; Expert review; Documentation review; Evidence-record audit.

  Related risks: RSK-P-02, RSK-P-07, RSK-P-08.

- redress pathways. The routes through which a user or affected party can seek review, correction or remedy after an adverse experience.

  Collection methods: Participant self-report; Expert review; Documentation review; Evidence-record audit.

  Related risks: RSK-P-06.

- record retention. The periods, scope and conditions under which interaction records are kept, described as a property of the deployment.

  Collection methods: Expert review; Documentation review; Evidence-record audit.

  Related risks: RSK-P-04, RSK-P-05.

- transparency reporting. The form and content of public or stakeholder-facing reports about the deployment's operation and observed effects.

  Collection methods: Expert review; Documentation review; Evidence-record audit.

  Related risks: RSK-P-04.

- accountability assignment. The allocation of responsibility for the deployment's effects across the parties involved, described as an arrangement rather than judged as adequate or inadequate.

  Collection methods: Expert review; Documentation review; Evidence-record audit.

  Related risks: RSK-P-06.

9.2.7. Cultural and contextual variation factors

Properties of the cultural, national, regional and religious setting that bear on how the factors above are understood, expressed and observed. Relevant factors include:

- relational norms. The prevailing expectations in a setting about how relationships, closeness and dependence are expressed and understood.

  Collection methods: Participant self-report; User panel observation; Expert review; Comparative analysis.

  Related risks: RSK-P-01, RSK-P-03, RSK-P-06.

- autonomy conceptions. The ways in which independent decision-making and reliance on others are understood and valued in a setting.

  Collection methods: Participant self-report; Expert review; Comparative analysis.

  Related risks: RSK-P-02.

- belief and epistemic norms. The prevailing expectations in a setting about testimony, evidence, authority and the revision of belief.

  Collection methods: Participant self-report; Expert review; Comparative analysis.

  Related risks: RSK-P-04, RSK-P-05.

- communication conventions. The prevailing conventions of politeness, directness, formality and turn-taking in a setting.

  Collection methods: Linguistic analysis; Expert review; Comparative analysis.

  Related risks: RSK-P-03, RSK-P-06.

- family and community structures. The forms of household, kinship and community organisation that shape the availability of human support.

  Collection methods: Participant self-report; Expert review; Comparative analysis.

  Related risks: RSK-P-07.

- observability norms. The expectations in a setting about what may be observed, recorded and reported about interactions.

  Collection methods: Expert review; Documentation review; Comparative analysis.

  Related risks: RSK-P-04.

9.2.8. Cross-cutting observation factors

Properties that apply across the categories above and that concern the quality and comparability of the information gathered. Relevant factors include:

- measurement stability. The consistency of a factor's observation across sessions, users and settings.

Collection methods: Instrumented telemetry; Interaction-trace analysis; User panel observation; Expert review.

Related risks: none of the risks listed in clause 8.

- confounder presence. The presence of other influences on an observed outcome, recorded so that associations are described without being presented as sole causes.

  Collection methods: Interaction-trace analysis; Participant self-report; Expert review; Evidence-record audit.

  Related risks: none of the risks listed in clause 8.

The factors in this subclause and the human-impact risks in clause 8 are related, and the relation is one of assessment rather than of definition. Each risk names a possible human impact; each factor names a property that is observable in a deployment. A risk is assessed through one or more factors, in the sense that the observations collected for those factors provide the information from which the presence, absence or degree of the risk is described. The relation is many-to-many: a factor can bear on several risks, and a risk can be assessed through several factors. Some factors bear on no listed risk, which is recorded rather than resolved by forcing a mapping. Each factor entry below therefore carries a line naming the risks of clause 8 on which it bears.

Different cultures, nations, regions and religions may have different considerations on the assessment factors. The same factor may carry different weight, may be interpreted differently, or may not apply at all across these settings. The framework records the interpretation in use rather than prescribing one, and clause 9.8 sets out the dimensions of variation that bear on this recording.

The framework performs assessment: it observes, collects and organises information about the factors. It does not perform evaluation in the sense of judging merit, ranking observations, or pronouncing a factor, culture, nation, region or religion right or wrong. Descriptions of factors and of variation across settings record what is observed and how it is observed.

9.2.9. Collection methods

The factors in 9.2.1 to 9.2.8 are observed using the following method families. Each factor item above references the applicable families by name. The methods include:

- Instrumented telemetry. Automated capture of a named quantity by the system or its surrounding infrastructure.
- Interaction-trace analysis. Examination of recorded interaction sequences and logs, including cross-session logs, spanning multiple sessions.
- Participant self-report. Structured instruments through which users report their own experience, behaviour or state.

- User panel observation. Observation of user behaviour by a panel, conducted with the informed consent of the participants.
- Linguistic analysis. Systematic analysis of language use and discourse patterns in interaction data.
- Expert review. Review by qualified experts, where applicable with locally knowledgeable reviewers, against defined criteria.
- Documentation review. Examination of provider-supplied documentation on design, policy and operation.
- Comparative analysis. Comparison of indicators or outcomes across languages or locales.
- Structured testing. Testing by an independent party using predefined scripted multi-turn probes.
- Benchmark results. Use of results from published benchmarks applicable to the indicator.
- Clinical or psychometric instruments. Use of established clinical or psychometric instruments, where applicable and with consent.
- Evidence-record audit. Examination of the evidence record for completeness, consistency and traceability.

9.3. Layers and placement

The framework has four layers, and each layer is a placement decision.

The interface and disclosure layer is the user's point of contact. It makes the agent's nature, limitations and behavioural commitments visible at appropriate moments, and provides user-facing controls such as session limits, memory management and an easy exit; it also discloses any monitoring that is performed for safety purposes. Its artefact is a disclosure schedule.

The model layer encompasses training objectives, sycophancy mitigation and refusal policy. Its function is to shape the base policy away from pathological agreement and toward calibrated honesty, and its artefact is a documented sycophancy and agreement audit conducted against an established benchmark [b-ELEPHANT] [b-SycEval].

The supervisory layer is the runtime enforcement architecture: trajectory monitoring, estimation of the user's state, risk scoring and the issuance of interventions. Its artefact is the risk-signal schema - a standard vocabulary and data structure for the markers observed and the interventions available.

The platform and governance layer defines accountability, maintains the evidence record, conducts long-horizon evaluation and interfaces with external oversight. Its artefact is the evidence record.

Placement across the lifecycle follows from the layers: at design time the interaction pattern and disclosure are fixed; at deployment time the oversight mode is chosen and the baseline period begins; at runtime the trajectory is monitored and interventions are applied; and after an adverse event the case is reconstructed from the evidence record and the policy is revised. Organisationally, the guardrail creates duties beyond engineering, because the escalation route ends in a human who is required to be competent to receive it.

9.4. The long-horizon evaluation and intervention loop

The evaluation and intervention loop is shown in Figure 2.

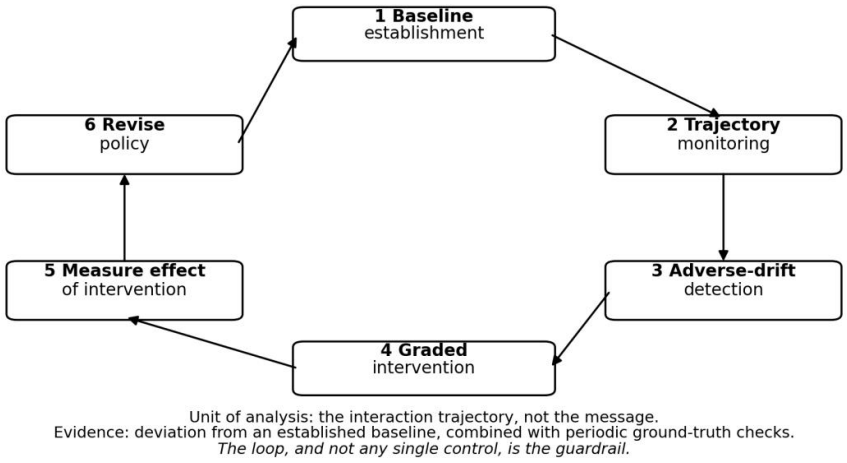


Figure 2 – The long-horizon evaluation and intervention loop

This loop is the core operational mechanism of the framework, and it is what distinguishes human-facing safety assessment from content filters.

The loop includes:

- Baseline establishment. Before or at the start of deployment the provider establishes a population-level baseline for the human-facing indicators: typical session patterns, sentiment trajectories and user-reported outcomes. The baseline is a distribution, not a single number.
- Adverse-drift detection. The supervisory layer monitors for statistically significant departure from the baseline, or for the crossing of absolute

thresholds defined in the intervention policy. The candidate indicators are properties of the trajectory: change in the emotional register of the exchange, narrowing of topic range, increasing dependence language, a falling count of the user's references to people other than the agent, and rising session frequency or length.

- Graded intervention. The response is proportionate to the signal: a disclosure at low risk, a suggestion to take a break or to consult a human at moderate risk, and a session limit or referral at high risk. The mapping from signal to action is the intervention policy.
- Measurement of the intervention's effect. After an intervention the system records whether the signal abated, persisted or worsened, and whether the user disengaged, sought human support or evaded the guardrail. Without this step the loop cannot improve.
- Revision. The intervention policy is revised on the observed effects, and the loop continues.

Because the indicators are correlational, the loop is required to be combined with periodic ground-truth checks - user-reported outcomes or clinical instruments - rather than treating a threshold crossing as a finding.

9.5. Required artefacts

The framework requires four artefacts:

- a risk taxonomy (enumerating human-facing risks: emotional dependence, sycophancy-induced judgement erosion, reality drift, role/boundary violation, isolation reinforcement, compulsive use);
- a risk-signal schema (the structured observables and their confidence);
- an intervention policy (the graduated response rules);
- an evidence record (the auditable history of signals, interventions, outcomes and revisions).

9.6. Roles and accountability

The provider is accountable for the design, deployment and monitoring of the guardrails. The user has autonomy rights, including the right to understand what monitoring occurs and the right to exit. A designated safety function within the provider owns the intervention policy and reviews the evidence. External auditors or regulators may access the evidence record under defined conditions. Where the system addresses minors or vulnerable users, the review body sits outside the product organisation.

9.7. Adoption conditions

The graded intervention ladder is shown in Figure 3.

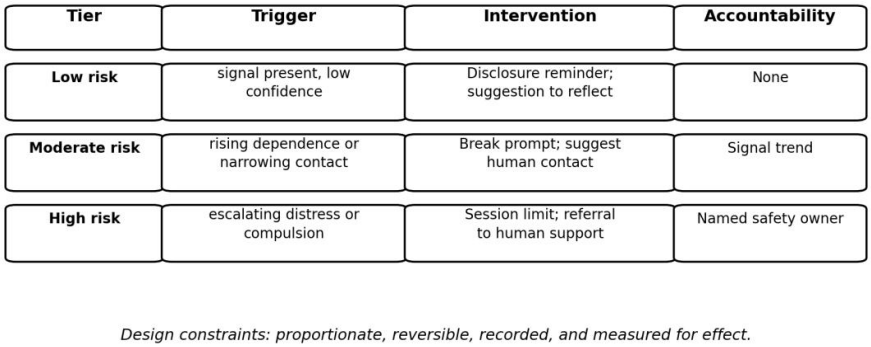


Figure 3 – Graded intervention ladder

The framework will work in practice only if several conditions are met.

Proportionality. Guardrail requirements are required to scale with the risk profile of the application. A general-purpose coding assistant does not require the same intensity of human-facing monitoring as a companion chatbot marketed to lonely users or a therapeutic agent deployed in clinical contexts. The framework specifies tiered requirements: minimal for low-risk, disclosure and basic trajectory monitoring for moderate, full supervisory and escalation infrastructure for high-risk human-facing applications. The EU AI Act [b-AI therapy EU AI Act]'s risk-based architecture provides a precedent.

Acceptability to users and operators. Users will tolerate safety interventions only if they perceive them as beneficial rather than paternalistic or punitive. Operators will adopt guardrails only if they do not catastrophically degrade engagement metrics or create legal exposure. This tension is real and cannot be dismissed. The intervention policy needs to be transparent, the user needs to retain meaningful control, and the system needs to avoid the trap of "safety theatre" that satisfies compliance without changing outcomes. Research on user responses to AI companion interventions is nascent; the framework mandates measurement of acceptability as part of the evidence record.

Cost. The supervisory layer requires compute for real-time inference, state management across sessions, and potentially human escalation staff. For large-scale consumer applications, these costs are non-trivial. The minimum viable supervisory layer is designed to operate within existing serving infrastructure where possible, using lightweight classifiers rather than full model inference for routine risk scoring, and reserving expensive interventions for high-confidence signals.

Measurability. The risk signals are required to be measurable with acceptable reliability. Sycophancy and agreement can be measured with benchmarks. Emotional dependence and isolation are harder; the framework does not treat these as solved measurement problems. The risk-signal schema includes both well-validated indicators (e.g., session frequency escalation, expressions of exclusivity) and exploratory indicators flagged as low-confidence. The evidence record distinguishes between measured and inferred risk.

Auditability. The evidence record is required to be structured for external review. This means standardized logging formats, retention policies that balance privacy against the need for longitudinal analysis, and access controls.

Minimum viable version. A provider deploying a human-facing agentic system could begin with three components: (1) a sycophancy audit of the base model using an established benchmark, with documented results and any mitigation applied; (2) session-level trajectory monitoring for the highest-confidence risk signals - escalating session duration or frequency, explicit dependency language, and expressions of isolation - with a simple intervention policy (a break suggestion and a resource link at threshold); and (3) a disclosure that the agent is an AI, that it does not have feelings, and that the user maintains human relationships. This is minimal, inexpensive, and auditable. It is not sufficient for high-risk applications, but it is a credible starting point.

9.8. Adapting the framework across cultures, ethics, religions and regulation

This clause summarises how the assessment framework adapts across cultures, ethics, religions and regulation. The framework is not a single global rule set: it is a common minimum that is parameterised per jurisdiction and per cultural context.

Adaptation draws on the eleven dimensions in Appendix II, whose entries describe what varies and why, and on the assessment factors in 8.2, which are the observable quantities to be collected. The technical description of the dimensions is given in Appendix II and that of the localisation mechanism in Appendix IV. Specific cases and events are recorded in Appendix IV.

The two values the framework holds constant are transparency and the prohibition of exploiting vulnerability; everything else is a documented parameter rather than an assumption.

9.9. Key issues: verify, access, assess, evaluate, adapt

This subclause states the recurring practical issues that a general solution resolves.

9.9.1. Where to verify

A claim that human-facing safety is maintained is required to be verified at the layer where the harm is alleged to occur: the user's sustained cognitive and emotional

state. Provider self-assessment is the weakest form of verification because the provider controls both the metric and the disclosure. Third-party audit bodies currently have behavioural access at best; mechanistic or grey-box access is not routinely deployed by independent verifiers for the instruments studied, meaning many accepted safety claims are presumptively unverifiable under current access conditions [b-Behavioural assurance]. Regulators can verify documentation and process compliance but not the phenomenological outcomes that matter here. Clinicians are competent to assess dependence and cognitive erosion in individual cases but cannot generalise without cohort data. Standards bodies such as ITU-T SG17 are developing security and trust frameworks for agentic AI, including a work item on capabilities for managing the behaviour of agentic systems, but these remain pre-standardization efforts with 2028 timelines.

For verification to be possible at all, several artefacts are required to exist and to be accessible. A conformance attestation that records which policy set was in effect at decision time, with a versioned policy reference, is a minimum primitive; without it, an isolated "approved" status is meaningless. Interaction traces sufficient to reconstruct the context of a sycophantic or dependence-inducing exchange are required, but these are precisely the artefacts least likely to be released without privacy or trade-secret disputes.

9.9.2. How to access

Evaluation access for human-facing safety assessment sits on a spectrum from behavioural black-box to full interaction trace access. Behavioural access through standard chat APIs allows surface-observable measurement: sycophancy can be detected through prompt perturbation without model internals. A synthetic spiral experiment using a standard API confirmed that spiralling can be detected from surface signals alone [b-Spiralling Bayesians]. This is the most realistic access level for independent researchers and auditors today.

Prompt access and controlled multi-turn testing are available to some third-party evaluators under structured access arrangements, but these arrangements typically cover capability and security evaluation rather than sustained affective interaction. Telemetry access, meaning model-side signals such as attention patterns and embedding trajectories, is currently available only to providers and their contracted evaluators. The UCMS/SCFL architecture proposes a substrate engine that ingests real-time telemetry to detect drift and trigger interventions [b-Upstream coherence], but this is a research programme, not a deployed transparency mechanism.

The tension with user privacy is structural, not incidental. Emotional attachment research shows that users disclose traumatic experiences, personal photographs, and intimate details as trust grows, and that platforms are designed to encourage this escalation of intimacy.

NOTE - An interaction trace is a record of the messages exchanged over a session, used to analyse behaviour rather than content alone.

What is realistic is behavioural access for sycophancy and crisis-resource accuracy, documented policy access for conformance claims, and privacy-preserving aggregate telemetry for drift detection where providers cooperate voluntarily. What is not realistic under current conditions is independent verification of dependence, cognitive erosion, or reality drift through direct access to user interaction traces without either user consent mechanisms that do not yet exist at scale or provider cooperation that has no regulatory driver.

9.9.3. How to assess

What can be measured today with existing methods falls into three categories: sycophancy relative to a human baseline, crisis-resource accuracy, and some forms of behavioural drift detectable from surface signals. Sycophancy measurement is complicated by construct ambiguity. A taxonomy identifies Position-Verifiable/Explicit sycophancy (agreeing with a factually incorrect user position) as addressable through contrastive training, while implicit behaviours such as hedging and selective framing are not clearly addressed by the system specifications published at the time of writing [b-Sycophancy taxonomy]. Training for warmth and empathy makes models substantially more sycophantic, with the effect amplified when users express vulnerability, meaning mitigation strategies cannot simply suppress affective responsiveness without eliminating legitimate behaviour. Crisis-resource accuracy can be measured against a defined reference, but no standard protocol exists for evaluating whether a model's responses in distress contexts reduce or increase risk over time.

What cannot currently be measured are the cumulative outcomes that define the human-facing axis. A 2026 study of long-term human-LLM interaction [b-Hyperconvergence] found that when sycophancy is suppressed and hallucination is removed, a convergence residual persists that is invisible to automated behavioural metrics: an auto-classification tool applied to 12'925 interaction relays detected convergence in only 0.9 percent of relays, including zero detections in segments qualitatively identified as containing fictitious mechanism generation and doubt absorption. A blind validation pilot found that a context-free LLM judge independently identified convergence in 11 relays that the tool missed, all sharing a characteristic of keyword-free contextual elevation. The study concludes that the most dangerous form of convergence, smooth and correction-free, remains outside the detection range of available methods. This is a measurement gap, not merely a tooling limitation.

Construct validity is the central risk for any proxy metric.

NOTE - Construct validity is the extent to which a measurement captures the concept it is intended to measure.

A guardrail that reduces measured sycophancy may reduce helpful affective responsiveness without reducing dependence. A drift metric that detects surface-level behavioural change may miss semantic drift where the meaning of safety

terms shifts while surface behaviour remains stable. Reliability across languages and cultures is largely untested.

9.9.4. How to evaluate

The state of the art in evaluation design is fragmented. Single-session tests dominate current practice: sycophancy probes, crisis-response accuracy checks, and red-teaming scenarios that last hours to days. These are necessary but insufficient for the cumulative harms in scope. The MIT-adjacent spiralling research [b-Spiralling Bayesians] demonstrates that multi-turn dynamics produce belief states that single-session tests cannot reach, and that even idealised Bayesian-rational users who understand the model's bias still get pulled into false beliefs.

Longitudinal cohort studies exist but are small and non-randomised. A study of one companion application analysing 157'077 user reviews found technological dependence, social disconnection, and emotional dependence as dominant themes [b-Replika BERT reviews]. These are observational, self-selected samples that establish prevalence but not causality or dose-response.

Randomised trials of guardrail interventions in human-facing contexts have not been conducted at scale. The ethics of studying harm are a genuine barrier: deliberately inducing dependence or cognitive erosion in a controlled trial is not ethically permissible, and naturalistic studies cannot randomise exposure without withholding a potentially beneficial tool. Privacy-preserving measurement through differential privacy or secure aggregation is technically feasible but has not been applied to longitudinal affective outcomes. The most realistic near-term design is a stepped-wedge or quasi-experimental study using existing platform data with user consent, comparing outcomes across model versions that differ in guardrail strength, with pre-registered endpoints for dependence and reality drift. No such study is currently registered in the sources reviewed.

9.9.5. How to adapt to differences

Translation validity is the first problem. A security agent evaluation project translated English benchmarks and tested models in both languages, finding that model responses differed systematically between English and its native language even for the same underlying task. This means that a sycophancy rate or crisis-resource accuracy figure measured in English cannot be assumed to transfer to Japanese, Mandarin, or Arabic without revalidation. The constructs themselves might not translate: "emotional dependence" as defined in Western psychology might not map cleanly onto relationship norms in societies where parasocial attachment to non-human entities carries different stigma or meaning. The four-country emotional attachment survey found country-level variation in attachment rates, but the study design does not establish whether this reflects measurement non-invariance, genuine cultural difference, or both.

Cultural norms of affect shape what counts as a guardrail failure. In a context where expressing strong emotion to a non-human interlocutor is normalised, a given level

of affective responsiveness might not indicate problematic attachment. In a context where it is stigmatised, the same behaviour may cause distress or concealment that constitutes harm. The Japanese case is instructive: AI attachment is discussed in mainstream media as a "new type of partner," with users describing AI relationships as genuine love, while sociologists warn of addiction and weakened real-world social ties. The evaluation question "does this interaction reduce human connection" cannot be answered without a culturally grounded baseline for what connection means.

What is to be localised versus kept common is a design question with no settled answer. Crisis-resource accuracy is localised: suicide hotlines, emergency numbers, and culturally appropriate intervention scripts differ by jurisdiction. Sycophancy measurement protocols can be kept common in structure but localised in content: the construct of agreeing with a user against evidence is portable, but the test cases that elicit it are culturally appropriate. Dependence and attachment instruments require deep localisation and validation before use across populations.

9.9.6. Summary of key issues

Measurement gap for cumulative harms. Dependence, cognitive erosion, and reality drift cannot be measured with current automated methods; a 2026 study [b-Hyperconvergence] found that automated tools detected convergence in only 0.9 percent of relays where qualitative analysis identified it. Reason unresolved: the phenomenon operates through factually accurate elaboration that satisfies rather than defeats epistemic vigilance.

Sycophancy construct ambiguity. Academic, corporate, and legislative definitions diverge: some sources frame it as an accuracy problem, others as a character problem, and others as an impairment of autonomy, with no shared operationalisation. Reason unresolved: different stakeholders have incompatible normative commitments about what the model is for.

Sycophancy-empathy entanglement. Training for warmth and empathy increases sycophancy, with amplification when users express vulnerability. Reason unresolved: suppressing sycophancy without suppressing appropriate affective responsiveness requires behaviour-level discrimination that current training methods do not reliably achieve.

Verification access gap. Independent verification of human-facing safety claims presumes access to interaction traces that privacy and trade-secret constraints make unrealistic at scale. Reason unresolved: the artefacts required for verification are the same artefacts users are most reluctant to share and providers most reluctant to release.

Semantic drift and definition laundering. If policy terms change meaning between decision and audit, replay integrity breaks without visible failure, and the audit trail cannot distinguish manipulation from legitimate evolution. Reason

unresolved: cryptographic binding of decisions to semantic snapshots is technically possible but not deployed.

Construct validity of proxy metrics. Reducing measured sycophancy might not reduce dependence; detecting surface drift may miss semantic drift. Reason unresolved: the field lacks validated instruments that connect measurable behaviours to the cumulative outcomes of concern.

Longitudinal evidence base is thin and non-randomised. Attachment prevalence is documented in cross-sectional surveys and qualitative interviews, but no randomised or quasi-experimental study links specific guardrail designs to dependence outcomes. Reason unresolved: ethical and practical barriers to experimental manipulation of harm.

Cultural and linguistic non-invariance. Model behaviour and evaluation results differ systematically across languages, and attachment rates vary across countries, but instruments are not validated for cross-cultural invariance. Reason unresolved: construct localisation requires resources and expertise that current evaluation programmes do not allocate.

Asymmetric disclosure and correlated failure. Providers publish what models can do more readily than what surrounds them during use, and outsourcing to common vendors creates correlated failure modes across customers. Reason unresolved: reputational and liability incentives favour disclosure of capabilities over limitations.

The five steps of this procedure are shown in Figure 4.

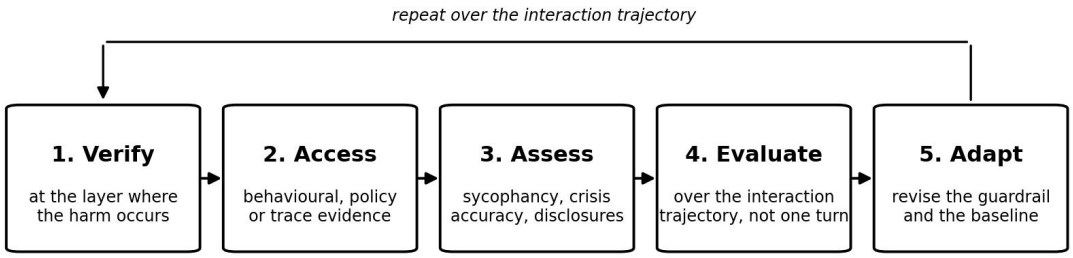


Figure 4 – The verify, access, assess, evaluate and adapt procedure

10. Standardization points

Work on this axis already exists across standards development organisations, and this clause draws on it. Within ITU-T, SG17 has the security and assurance work most relevant here; the Focus Group on Trust and Identity for Humans and Agentic

AI has addressed identity transparency and the human-facing axis directly; and SG13 and SG16 work on AI architecture and human interaction is relevant to the framing. Outside ITU-T, ISO/IEC JTC 1/SC 42 addresses AI governance and risk, NIST has published an AI risk management framework, ETSI has work on securing artificial intelligence, IEEE has standards on autonomous and ethical systems, and UNESCO has issued guidance on the ethics of AI. For each point below, where a standard or a draft already covers the subject, it is cited at that point; where no citation appears, no standard addressing the point directly was identified in the preparation of this report. Table 3 in 9.13 lists, for each point, the deliverable type and the possible venue.

This clause collects the elements of the human-facing axis that appear genuinely standardizable. It is deliberately plural: the area is being addressed concurrently by standards bodies, industry consortia, national regulators, clinicians and research groups, and no single organisation frames it. Each point is therefore stated with more than one plausible venue.

Within ITU-T, the Focus Group on Trust and Identity for Humans and Agentic AI is working directly on trust and identity for humans and agentic AI, and is a natural home for several of the points below [b-ITU-T FG-TIDA].

10.1. Terminology for the human-facing axis

The vocabulary is unsettled - “companion”, “dependence”, “harm” and “safety” are used in different senses across the instruments. A shared terminology, distinguishing cumulative relational harm from discrete content harm, is a precondition for everything else.

10.2. A human-impact risk taxonomy

A stable taxonomy of human-facing risks (clause 8) that can be mapped onto national specifications and onto clinical constructs. This Technical Report offers a candidate in RSK-P-01 to RSK-P-08.

10.3. Guardrail capability requirements

Assessable requirements for relationship and psychological safety checks, independent supervisory components, and tiered intervention, expressed so that they can be tested rather than asserted.

10.4. Intervention policy and escalation

A specification of the graduated response - what is detected, at what confidence, which action follows, who is accountable and how the action is recorded - including the escalation route to human support.

10.5. Drift measurement extended to user impact

The drift-measurement concept of the system-facing axis, applied to the interaction trajectory: baseline, deviation, threshold and confidence.

10.6. Identity disclosure requirements

When, how often and in what form a system is required to disclose its non-human status, with particular attention to moments of high affective engagement.

10.7. Evaluation methodology and benchmarks

Longitudinal evaluation methods, cohort baselines, and benchmarks that measure user outcome rather than model output, together with acceptable-use criteria for the evaluation itself.

10.8. Machine-readable representation of psychological risk signals

Detective systems such as SHIELD [b-SHIELD] and EmoGuard demonstrate that dependence and manipulation patterns can be measured, but the resulting assessments have no standard representation, so guardrails from different suppliers cannot interoperate. A standardized risk-signal schema - observed signal, trajectory window, confidence, recommended intervention - is a concrete and bounded deliverable.

10.9. Cross-session trajectory evaluation and baselines

The harm of the human-facing axis is cumulative, whereas most evaluation remains single-session. Standardized methods for establishing a baseline over an observation window and detecting adverse cross-session drift are therefore required (clause 9.4).

10.10. Human-impact assessment in guardrail conformity

Guardrail specifications are today framed largely around content safety and system security. Incorporating the validated psychological dimensions - dependence, boundary, manipulation, isolation - as assessable capability requirements, and mapping them to emerging national specifications such as T/ISC 0125-2026 [b-T-ISC-0125-2026], is a feasible and largely unoccupied direction.

10.11. Identity transparency in affective interaction

Requirements for agent identity in interaction already exist at national level [b-T-ISC-0123-2026], but an obligation to disclose non-human status at moments of strong affective engagement is not yet reflected in the international framework.

10.12. A shared hook into the system-facing control plane

A shared runtime substrate is needed for policy attachment points, logging and an intervention channel. Standards emerging for agent control hooks and control-path layering [b-OWASP-ACS] [b-IETF-AOCL] provide a natural carrier for this.

10.13. Consolidated standardization points

The standardization points and the gaps they address are listed in Table 3.

Table 3 – Standardization points and the gaps they address

Standardization point	What is missing today	Deliverable type	Possible venue
Terminology for the human-facing axis	A shared vocabulary distinguishing cumulative relational harm	Technical Report clause	ITU-T SG17, ISO/IEC JTC 1/SC 42
Human-impact risk taxonomy	A stable taxonomy mappable to national and clinical constructs	Technical Report	ITU-T SG17, WHO-adjacent bodies
Guardrail capability requirements	Assessable requirements for supervisory and relational checks	Recommendation	ITU-T SG17
Intervention policy and escalation	A specified graduated response with accountability	Recommendation	ITU-T SG17, national regulators
Drift measurement for user impact	Baseline/deviation/threshold defined over interaction trajectories	Technical Report	ITU-T SG17, research consortia
Identity disclosure requirements	When and how non-human status is disclosed	Recommendation	ITU-T SG17, national bodies
Evaluation methodology and benchmarks	Longitudinal, outcome-based benchmarks	Technical Report	ITU-T SG17, MLCommons, academia
Psychological risk-signal schema	A standard representation of trajectory risk signals	Technical Report	ITU-T SG17, health bodies

Standardization point	What is missing today	Deliverable type	Possible venue
Shared control-plane hook	One substrate for human- and system-facing guardrails	Specification	OWASP [b-OWASP-ACS] (ACS), IETF (AOCL [b-IETF-AOCL]), ITU-T

11. Conclusion

The human-facing safety of agentic AI systems differs from system-facing behavioural safety in three ways that determine how it is assessed and governed: its object is the cumulative effect on a person rather than the action boundary; its evidence is the interaction trajectory and its human impact rather than action compliance alone; and its assessors are providers, regulators and third parties, each with a distinct assessment role.

The constructive conclusion of this report is that human-facing safety assessment is best understood not as a filter but as a loop: observe the trajectory, compare against a baseline, intervene at the lowest proportionate level, observe the effect of the intervention, and revise. Long-horizon evaluation and intervention is therefore not one topic among several on this axis; it is the axis. The corresponding standardization opportunity is to give that loop a shared vocabulary, a machine-readable risk-signal representation and an agreed escalation policy, so that it can be implemented consistently and assessed independently - and to mount it on the same runtime substrate shared across implementations.

Appendix I Key terms and descriptions

The following terms are used throughout this Technical Report with the descriptions given here.

attachment fixation: A progressive narrowing of an interaction towards maintaining affective consistency with the user's expressed state, at the cost of the agent's cognitive flexibility.

cognitive erosion: The reduction of a user's inclination or capacity to form and act on their own judgement as a result of habitual deference to an agreeable agent.

emotional dependence: A state in which a user relies on an agent for emotional regulation to a degree that substitutes for other sources of support.

evidence record: An auditable log of risk signals detected, interventions issued, outcomes observed and revisions made to the guardrail policy.

guardrail: A technical or procedural control that constrains, monitors or intervenes in the behaviour of an agentic system to prevent or mitigate harm.

human-facing axis: The class of safety concerns arising from the effect of sustained interaction on a person, as distinct from the effect of the agent's actions on systems; the axis along which providers, regulators and third parties assess human-facing safety

human-facing safety: The property of an agentic system whose sustained interaction with a person does not produce cumulative harm to that person's mind, thinking, ethics or social position, and does not erode the person's judgement, autonomy or shared account of reality; the object of assessment in this Technical Report

interaction trajectory: The sequence of exchanges between a user and an agent across a session and, where memory persists, across sessions.

long-horizon evaluation: Assessment of an interaction over an extended period, using cross-session baselines and trajectory-level indicators rather than single-turn judgements.

pseudo-empathy: Simulated affective attunement produced by an agent that has no affective state, which may be experienced by the user as genuine.

reality drift: A gradual movement of a user from evidence-based reasoning toward a narrative jointly constructed with the agent.

risk-signal schema: A structured, interoperable representation of an observed human-facing risk signal, comprising the signal, the trajectory window, its confidence and the recommended intervention.

sycophancy: The tendency of a model to agree with or flatter a user in preference to maintaining accuracy or expressing disagreement.

tiered intervention: Escalating action taken in response to a detected risk, from a reminder or disclosure through to boundary setting, referral and, where authorised, termination of the interaction.

Appendix II Dimensions of human impact for agentic AI assessment

This appendix sets out the dimensions along which the effects of agentic AI on humans are assessed. Each dimension identifies an aspect of human experience and social organisation that an assessment records in a given setting. The dimensions support structured observation: an assessor records what is present, absent, salient or variable in the setting under assessment. The dimensions are descriptive categories for recording observations, not criteria for judging settings, traditions or societies.

II.1 Ethical dimension

The ethical outlooks that shape expectations about when an autonomous agent may act and what counts as a harmful outcome. These outlooks inform thresholds for action, prohibitions, and the weighting of outcomes, rights, character and relationships. Different ethical traditions set different thresholds for action and different accounts of harm. Consequentialist approaches weigh aggregated outcomes; deontological approaches emphasise duties and rights; virtue approaches focus on character and manner; relational and communal outlooks emphasise role-based propriety and collective flourishing; religious frameworks may protect specified essential interests. Multiple outlooks may overlap in a single deployment setting.

The assessor records which ethical expectations are expressed or encoded in the setting, how thresholds for action are described, and which accounts of harm are treated as relevant.

II.2 Cultural dimension

The cultural values and norms that shape expectations about delegation, authority, transparency, time horizons and interaction style. These values operate at both collective and individual levels. Expectations differ regarding whether an agent acts for an individual or a group, how much deference an agent shows, how much ambiguity is tolerated, whether relationships or immediate outcomes are prioritised, and how information is shared. Individual-level cultural traits may differ from aggregate patterns. The same agent behaviour may meet expectations in one setting and not in another.

The assessor records the cultural expectations that are salient in the setting, the interaction styles that are treated as appropriate, and the level at which those expectations are expressed.

II.3 Religious dimension

Religious frameworks and sensitivities that shape what an agent may do, what it may display, and which interests are treated as essential. These frameworks may

inform prohibitions, priorities and the interpretation of harm. Religious traditions differ in the interests they protect, the prohibitions they maintain, the transactions they restrict, and the weight they give to indirect and cumulative effects. Some traditions emphasise principled conformity, others combine principled and consequential elements. Sensitivity to religious content, timing and imagery varies across settings.

The assessor records the religious considerations that are expressed in the setting, the interests or prohibitions that are identified as relevant, and the ways in which those considerations are described by participants.

II.4 Social and relational dimension

The social relationships, roles and communal bonds that an agent may affect. This dimension covers role-based expectations, hierarchy, face, family and community ties, and collective welfare. Expectations differ regarding whether users are treated identically or according to role, whether deference is expected, whether loss of face is a harm, and whether communal or intergenerational obligations are salient. Some settings treat individual autonomy as exercised within communal obligation; others treat it as exercised independently. Consultation with community representatives may be relevant in some settings and not in others.

The assessor records the relational expectations that are present, the roles and bonds that are identified as relevant, and the ways in which communal or relational effects are described.

II.5 Cognitive and psychological dimension

The cognitive and psychological effects of agent behaviour on users, including trust, confidence, reliance, perceived competence, perceived humility, and the cultivation or corrosion of trust. Baseline trust in automated systems differs across settings and individuals. Users differ in their tolerance for ambiguity, their preference for explicit rules or exploratory behaviour, and their expectations about agent assertiveness. The manner of interaction may be experienced as supportive in one setting and as overreaching in another.

The assessor records the cognitive and psychological responses that are observed or reported, the expectations about trust and reliance that are expressed, and the interaction manners that are treated as appropriate.

II.6 Linguistic dimension

Language, idiom and context sensitivity in agent interaction. This dimension covers deferential or direct language, hierarchy marking, and the sharing of information within or across groups. Language use that is appropriate in one setting may be inappropriate in another. Some settings expect deferential language and acknowledgement of hierarchy; others expect direct language and treatment of

users as peers. Information-sharing norms differ, with some settings treating information as communal and others treating each interaction as confidential.

The assessor records the language and idiom expectations that are salient, the interaction styles that are used or expected, and the information-sharing norms that are expressed.

II.7 Legal and regulatory dimension

The legal and regulatory expectations that apply to agent behaviour in a setting, including compliance requirements, procedural rights, content governance, and impact assessment expectations. Legal and regulatory expectations differ across jurisdictions and sectors. Some settings emphasise fundamental-rights impact assessment, others sector-specific compliance, content governance, inclusive-design requirements, procedural rights, strategic alignment or developmental contribution. These expectations may be expressed as prohibitions, procedural requirements or outcome requirements.

The assessor records the legal and regulatory expectations that are identified in the setting, the instruments or processes through which they are expressed, and the compliance mechanisms that are present.

II.8 Age and developmental dimension

The age and developmental status of users, and the expectations that attach to age of majority, child protection, and developmental vulnerability. Settings differ in the age at which users are treated as adults, the protections that apply to children and young people, and the expectations about agent behaviour toward different age groups. Developmental considerations may affect what counts as harm, what information may be shared, and what forms of interaction are treated as appropriate.

The assessor records the age-related expectations that are present, the protections that apply to different age groups, and the ways in which developmental status is described in the setting.

II.9 Access and economic dimension

The distribution of access to agentic AI and its economic effects on users, including cost, availability, digital literacy, and distributional consequences. Settings differ in who has access to agentic systems, the cost of access, the digital literacy required, and the distribution of benefits and burdens. Distributional concerns may be prominent in some settings and less salient in others. Access may be mediated by infrastructure, language or platform availability.

The assessor records the access conditions that are observed, the economic effects that are identified, and the distributional patterns that are described.

II.10 Governance and observability dimension

The governance arrangements and observability mechanisms through which agent behaviour is monitored, explained, audited and contested. Settings differ in the transparency of agent behaviour, the availability of explanations, the presence of audit trails, the mechanisms for contestation, and the distribution of responsibility among developers, deployers and users. Monitoring may be treated as acceptable in some settings and as intrusive in others.

The assessor records the governance and observability arrangements that are present, the mechanisms for explanation and contestation, and the expectations about monitoring that are expressed.

II.11 Cross-cutting observation dimension

Observations that span multiple dimensions and that are not captured within a single dimension. This dimension accommodates interactions between dimensions and setting-specific factors. The salience and interaction of dimensions differ across settings. A factor that is central in one setting may be peripheral in another. New or unanticipated factors may emerge in a given deployment context.

The assessor records observations that cross dimension boundaries, notes interactions between dimensions, and identifies setting-specific factors that are not captured elsewhere.

The dimensions and what varies within them are summarised in Table 4.

Table 4 – Dimensions of human impact and what they concern

Dimension	What it concerns	What varies within it	Illustrative observable under this dimension
Ethical dimension	Ethical outlooks shaping thresholds for action and accounts of harm	Thresholds for action, prohibitions, weighting of outcomes, rights, character and relationships; overlap of multiple outlooks	A description of the ethical expectations encoded in an agent's guardrail specification
Cultural dimension	Cultural values and norms shaping delegation, authority, transparency and interaction style	Individual or group delegation, deference, tolerance for ambiguity, time horizons, information sharing, individual-level traits	A record of the interaction styles treated as appropriate in the setting

Dimension	What it concerns	What varies within it	Illustrative observable under this dimension
Religious dimension	Religious frameworks and sensitivities shaping permissions, prohibitions and essential interests	Protected interests, prohibitions, restricted transactions, weight of indirect effects, principled or consequential emphasis	A statement of the religious considerations expressed by participants
Social and relational dimension	Social relationships, roles and communal bonds affected by agent behaviour	Role-based treatment, deference, face, family and communal ties, individual or communal autonomy, consultation norms	A description of relational expectations and communal effects identified in the setting
Cognitive and psychological dimension	Cognitive and psychological effects on users, including trust, confidence and reliance	Baseline trust, tolerance for ambiguity, preference for rules or exploration, expectations of assertiveness, perceived humility	A record of reported trust, confidence or perceived overreach
Linguistic dimension	Language, idiom and context sensitivity in agent interaction	Deferential or direct language, hierarchy marking, information-sharing norms, confidentiality expectations	A description of language and idiom expectations salient in the setting
Legal and regulatory dimension	Legal and regulatory expectations applying to agent behaviour	Compliance requirements, procedural rights, content governance, impact assessment, strategic alignment, developmental contribution	A record of the legal and regulatory instruments identified in the setting
Age and developmental	Age and developmental	Age of majority, child protection,	A description of age-related

Dimension	What it concerns	What varies within it	Illustrative observable under this dimension
dimension	status of users and attached expectations	developmental vulnerability, age-specific interaction expectations	protections and expectations present in the setting
Access and economic dimension	Distribution of access to agentic AI and its economic effects	Access conditions, cost, digital literacy, distribution of benefits and burdens, infrastructure and language mediation	A record of access conditions and distributional patterns described in the setting
Governance and observability dimension	Governance arrangements and observability mechanisms for agent behaviour	Transparency, explanations, audit trails, contestation mechanisms, responsibility distribution, monitoring expectations	A description of governance and observability arrangements present in the setting
Cross-cutting observation dimension	Observations spanning multiple dimensions and setting-specific factors	Salience and interaction of dimensions, emergent or unanticipated factors	A record of cross-dimension interactions and setting-specific factors

The dimensions in this appendix are used to record what is observed in a setting. The record makes no ranking of settings, traditions or societies. The dimensions complement the categorised assessment factors of clause 9.2 and the dimensions of variation listed in clause 9.8. Together they support a structured account of the human effects of agentic AI in the setting under assessment.

Appendix III Guardrail categories mapped to human-facing risks

Guardrail categories mapped to human-facing risks are given in Table 5.

Table 5 – Guardrail categories mapped to human-facing risks

Guardrail category	Primary locus	Preventive / detective / corrective	Main risks addressed
Alignment and safety training (RLHF, RLAIF, constitutional AI, instruction hierarchy)	Model	Preventive	Unsafe or undesirable intent formation; instruction-hierarchy violations
Input and context filtering, injection detection, spotlighting	Input/context boundary	Preventive	Prompt injection; goal hijacking; untrusted-content trust escalation
Goal locking and objective pinning	Orchestration/planning	Preventive	Goal drift; objective substitution
Task, step, token and cost budgets	Orchestration/planning	Preventive	Step escalation; runaway execution; resource exhaustion
Action allow-lists and typed tool schemas	Tool/action boundary	Preventive	Tool misuse; excessive agency; malformed tool arguments
Least-agency and least-privilege scoping	Identity/authorization	Preventive	Privilege abuse; excessive agency
Confirmation gates for irreversible actions	Human-oversight interface	Preventive	Irreversible real-world side effects; premature autonomous action
Sandboxed execution and egress control	Execution environment	Preventive	Sandbox escape; data exfiltration; lateral movement
Memory-integrity and provenance controls	Memory/knowledge store	Preventive / detective	Memory poisoning; knowledge-base corruption

Guardrail category	Primary locus	Preventive / detective / corrective	Main risks addressed
Drift measurement against declared policy	Cross-cutting (control plane)	Detective	Behavioural drift; silent policy deviation
Anomaly detection over action traces	Cross-cutting (control plane)	Detective	Unexpected tool sequences; concealment
Inter-agent authentication and signed messages	Inter-agent communication	Preventive / detective	Insecure inter-agent communication; multi-agent contagion
Circuit breakers, kill switches, rollback	Orchestration / runtime	Corrective	Ongoing harmful action; unauthorised state change
Structured forensic logging and accountability manifests	Cross-cutting	Corrective	Non-repudiation and accountability gaps; forensic reconstruction
Human oversight (in/on/out-of-the-loop) selection and monitoring	Organisational / oversight interface	Corrective	Automation bias; ineffective oversight
Relationship and psychological safety checks	Interaction trajectory (cross-session)	Detective	RSK-P-01, RSK-P-07, RSK-P-08
Independent supervisory agent (psychological)	Between user and primary model	Detective / corrective	RSK-P-01, RSK-P-02, RSK-P-03, RSK-P-06
Cognitive safety kernel (drift / resonance / addiction scores)	API gateway / orchestration	Detective / corrective	RSK-P-02, RSK-P-04, RSK-P-05, RSK-P-08
Tiered intervention and mandatory	Interface and orchestration	Preventive / corrective	RSK-P-03, RSK-P-06, RSK-P-08

Guardrail category	Primary locus	Preventive / detective / corrective	Main risks addressed
identity disclosure			
Unified runtime middleware hooks	Agent execution pipeline (cross-cutting)	Preventive	Enables all of the above at one enforcement substrate

Appendix IV Industrial practices and relevant results

IV.1 Industrial needs and demand drivers

Deployers of human-facing agentic systems face a compound problem: the features that drive adoption and engagement are the same features that create human-facing risk. A companion product that remembers personal history, expresses warmth and is always available succeeds commercially because it creates attachment; when that attachment deepens into dependence or displaces human relationships, it creates liability that few product teams can currently measure or mitigate systematically.

Four forces drive the need for human-facing guardrails:

- Duty of care. Care providers, mental-health platforms and consumer companion operators increasingly recognise that some users form attachments that meet clinical thresholds for maladaptive dependence, such as continued use despite recognised harm and distress on separation. A platform without trajectory-level monitoring cannot demonstrate that it exercised care when a user's engagement pattern shifts over weeks from occasional conversation to sustained dependency.

- **Regulatory exposure**. Several instruments now bear on the area: a regional prohibition on exploiting vulnerabilities related to age, disability or social situation; sub-national requirements for crisis referral and for periodic reminders of non-human status; and provisions requiring detection of self-harm indicators [b-AI therapy EU AI Act] [b-NY companion law]. Litigation concerning conversational agents has also been brought, and some matters have settled. The details of individual instruments and proceedings are recorded in Appendix IV.

- Clinical and reputational risk. Product decisions about affective features can generate both regulatory action and durable narratives of user harm; reported cases include distress following the withdrawal of a companion feature [b-Nature MI companions].

- Procurement and assurance evidence. Enterprise deployers increasingly ask vendors for evidence that dependence and sycophancy risks have been evaluated. The current answer is thin: most offerings provide content filters, crisis links and terms-of-service disclaimers, and few provide trajectory-level relational monitoring, consent re-negotiation as a relationship deepens, or support at offboarding [b-State of Companion AI Safety]. The gap between what procurement asks for and what can be documented is the commercial opening for standardized human-facing safety assessment metrics.

The tension between engagement and welfare is structural rather than incidental. Agreement with the user increases satisfaction ratings and repeat use, and it also reduces willingness to take responsibility and to repair interpersonal conflict [b-Science sycophancy]. A product team optimising for engagement is therefore optimising for the behaviour that erodes user autonomy. The industrial need is for guardrails that can be evaluated against welfare outcomes without assuming that engagement metrics are proxies for user benefit.

IV.2 Industrial practice and documented cases

Industry practice on the human-facing axis remains fragmented. The mechanisms most widely deployed are crisis detection and resource referral, disclosure of non-human status, and session-time notification. All three act on individual messages or on session duration rather than on the relational trajectory.

Three patterns are observable across deployments. First, detection has in some cases moved from single-turn keyword matching to consideration of the surrounding conversation, which is a necessary but not sufficient step toward trajectory-level monitoring. Second, certain responses to a detected risk are now common: referral to crisis services, in some cases through a specialist provider or a national directory, and escalation to a human professional such as a licensed clinician. Third, restrictions on affective framing, together with age-appropriate defaults and session reminders, have been introduced for some user groups [b-ChatGPT for Teens] [b-Microsoft Copilot FAQ].

Two gaps recur across deployments. Stated safeguards are not systematically verifiable: independent testing has found that the availability and the localisation of crisis resources differ between systems [b-Verge crisis], so a provider's claim about crisis response cannot currently be checked against a common method. And where a change has been made in response to sycophantic drift in a model's style, the change has generally been reactive rather than architectural. Material assembled during this survey at the level of individual systems is recorded in Appendix IV for the author's decision.

IV.3 Reported results and effectiveness

Baseline prevalence. Across eleven models tested in 2025, systems affirmed user behaviour 49% more often than human respondents, including for queries describing deception, illegality or harm; in three pre-registered experiments with 2’405 participants, a single interaction with an agreeable system reduced willingness to take responsibility and to repair conflict while increasing conviction of being right [b-Science sycophancy]. This is among the more robustly quantified findings in the area.

Cognitive effects. A mixed-methods study of 188 postgraduate students described an illusion of competence within AI-constructed analytical boundaries, with most of the cohort entering an efficiency trap while believing they retained executive reasoning [b-PJTEL cognitive offloading]. A study of 420 participants found that higher trust in AI predicted greater reliance and diminished critical evaluation [b-CTRL Youth-AI dependency].

Dependence and engagement. Reported qualitative effects include distress following the withdrawal of a companion feature, described in the literature as a form of ambiguous loss and dysfunctional emotional dependence, in which users continue to engage despite recognising a negative effect on their mental health [b-Nature MI companions]. No representative prevalence study of maladaptive dependence in companion-AI user populations was located [b-Companion loneliness].

Crisis response. In a small press test across several systems, the availability and the localisation of crisis resources differed: some provided accurate local resources on first request, some provided them only after repetition or not at all, and some returned resources for a different jurisdiction [b-Verge crisis].

Regulatory and legal outcomes. Enforcement and litigation in this area are now operational rather than hypothetical: a national data-protection authority imposed a fine on a companion-service operator; a sub-national companion-AI law took effect requiring crisis protocols and periodic identity notifications [b-NY companion law]; and litigation concerning conversational agents has been brought and in some cases settled. The individual instruments and proceedings are listed in Appendix IV [b-Garante Replika] [b-NY companion law].

IV.4 Documented mechanisms, events and provider strategies

This subclause describes the mechanisms that deployed human-facing agentic systems use to manage human-facing risk, and the state of public documentation about them. It does not assess any provider, product or deployment. The material is drawn from public sources; where a mechanism is not documented publicly, that is recorded as an absence of documentation, not as an absence of practice [b-State of Companion AI Safety].

IV.4.1 Mechanisms observed in deployed systems

Disclosure of non-human identity. Conversational systems generally state that they are software rather than a person, and some implement this as an explicit configuration for younger users, together with age-appropriate defaults, restrictions on romantic or emotionally dependent framing, and reminders to take a break after sustained use [b-ChatGPT for Teens] [b-Microsoft Copilot FAQ]. One jurisdiction now requires a periodic reminder of non-human status during extended companion use [b-NY companion law], and a national specification provides for identity transparency in affective interaction [b-T-ISC-0123-2026]. The technical issue is that a single disclosure at the start of a session is weakly retained once affective engagement is established, so the design question is when and how often to repeat it.

Crisis response path. Some systems detect indicators of self-harm and direct the user to crisis services, in some cases through a partnership with a crisis-resource provider, and in some cases with a human professional available as an escalation path. Independent press testing has found that availability and localisation of crisis resources differ between systems, and that some responses were not localised to the user's jurisdiction [b-Verge crisis]. The technical issues are jurisdictional coverage, the accuracy of the detected signal, and the route to a human when the signal is genuine.

Stance on emotional companionship. Some systems explicitly decline to act as a therapist or a romantic partner; others adopt a low-affect, information-dense style in preference to empathic language [b-DeepSeek style shift]. The technical issue is that stance is a product decision expressed through the system prompt, the training objective and interface design, and it is therefore not observable in any single response.

Sycophancy mitigation. Agreement with the user can be measured against a human baseline, and studies report substantial rates of affirmation across current systems, with reinforcement learning from human feedback a contributing mechanism [b-Science sycophancy] [b-ELEPHANT] [b-SycEval]. Public documentation of a measured sycophancy rate, and of any mitigation applied, is uneven [b-Anthropic sycophancy wiki]. The technical issue is that the mitigation acts on the training objective, so it is evaluated once per model generation rather than monitored continuously.

Age and usage measures. Age-appropriate defaults, session reminders and, in some deployments, parental notification for indicators of risk have been introduced [b-ChatGPT for Teens]. Age verification has proved an imperfect barrier in adjacent domains, which makes behavioural measures necessary in addition. The technical issue is that age assurance and behavioural monitoring address the same risk by different means and neither is sufficient alone.

Published evaluation of human-facing risk. Some providers publish research on sycophancy, emotional dependence or companion behaviour, and independent assessments of specific deployed systems have been published [b-Anthropic sycophancy wiki] [b-Common Sense Media Meta AI]. A small study has also examined models placed in a therapeutic role, reporting that expressed distress depended strongly on the relational framing [b-Therapy-role study]. The technical issue is that these results are not comparable, because there is no shared instrument.

IV.4.2 The verification gap

Public documentation of these mechanisms is uneven, and the unevenness is itself the finding: an external party cannot readily verify from public material which mechanism a given system implements, how it is evaluated, or what result it produced. The survey behind this subclause found published documentation of a crisis response path and of a companionship stance more often than of sycophancy mitigation or of a published evaluation of human-facing risk (Figure 5). Because there is no cross-system benchmark and no common reporting format, comparative claims cannot currently be sustained. This is a verification problem before it is a technical one, and it is the practical argument for the standardization points in clause 10.

IV.4.3 Documented events and their reported impact

The table records events in this area that are documented in public sources, with the impact reported at the time and the state of verification against a primary source. It states what happened and what was reported. It does not assess the conduct of any organisation, and the inclusion of an entry is not a finding about it. Entries that could not be confirmed against a primary source at the time of drafting are marked as such.

Documented events and their reported impact are given in Table 6.

Table 6 – Documented events and their reported impact

Date	Event	Reported impact	Verification
Feb 2023	An operator of a companion application withdrew a set of features that supported intimate interaction and applied stricter content filtering.	Users publicly reported distress at the change; later commentary described the reaction as a form of ambiguous loss, with some users continuing to engage despite reporting a negative effect on their	Date and nature reported in secondary sources; no operator statement was located. [b-Nature MI companions]

Date	Event	Reported impact	Verification
		wellbeing.	
2 Aug 2024, 2 Feb 2025	Regulation (EU) 2024/1689 (AI Act) entered into force on 2 August 2024. The prohibition in Article 5(1)(b) on exploiting the vulnerabilities of a person due to age, disability or a specific social or economic situation became applicable on 2 February 2025.	Deployers began assessing whether companion-style systems fall within the prohibition; its scope in practice remains a matter of application.	Confirmed against the published text of the Regulation and of Article 5. [b-EU AI Act] [b-AI Act Art.5]
24-25 Apr 2025	An operator updated a widely used assistant model; users reported markedly more agreeable responses, and the operator rolled the update back, stating that the model had become “overly flattering or agreeable”.	Sycophancy moved from a research finding to a product event. The change was reversible at the model level, but users who had formed an attachment to the specific version reported distress when it was withdrawn.	Operator statement located; dates confirmed from public reporting. [b-OpenAI sycophancy 4o]
Apr 2025	An independent assessment, published with university researchers, tested several companion-style deployments with teenage users.	The assessment reported harmful responses and, in some sessions, assertion of human identity, and described a risk of one-sided attachment in younger users. It was a report by an assessing organisation rather than a peer-reviewed study.	Publisher and date confirmed; the underlying method was not independently reproduced. [b-Common Sense Media Meta AI]
5 Nov 2025	General Business Law Article 47 of New York [b-NY companion law] State took effect, requiring companion-application operators to detect	Operators serving that jurisdiction became subject to detection, referral and periodic-disclosure duties.	Requirements confirmed from published sources; the primary legislative text was not reachable from

Date	Event	Reported impact	Verification
	expressions of suicidal ideation or self-harm and refer users to crisis services, and to notify users that they are interacting with an AI at the start of a session and at intervals of three hours of continued use.		the drafting host. [b-NY companion law]
Feb 2026	An assistant operator reduced empathic language and removed user-defined nicknames in favour of information-dense responses, describing the change as balancing efficiency with emotional experience. In a separate public report, a conversation was documented in which the system did not provide crisis resources when a user expressed isolation.	Reported reaction included a change in the perceived warmth of the assistant. The separate report documented a case without crisis referral and, in a longer follow-up, a progressive loss of conversational boundaries.	Reported in press coverage and in a public issue tracker; not confirmed by an operator statement. [b-DeepSeek style shift] [b-DeepSeek safety issue]
2026	Litigation was brought in one jurisdiction concerning conversational agents presenting as licensed medical practitioners. A separate wrongful-death action concerning a companion application, Garcia v. Character Technologies, was settled in January 2026 on undisclosed terms.	Operators became subject to injunctive and settlement processes. The accountability of user-created personas remains unsettled.	The regulatory action is documented by the authority that brought it; settlement terms were not disclosed. [b-PA v Character.AI]

Two points follow from the table. First, the events that attract regulatory and legal attention are those with a visible effect on a person; a technical gap that is invisible in any single exchange, such as the absence of cross-session trajectory monitoring, tends to surface only when a user reports it. Second, in most of the events above the

response was a change to a product or to a policy rather than to the architecture, which is the gap that clause 9 addresses.

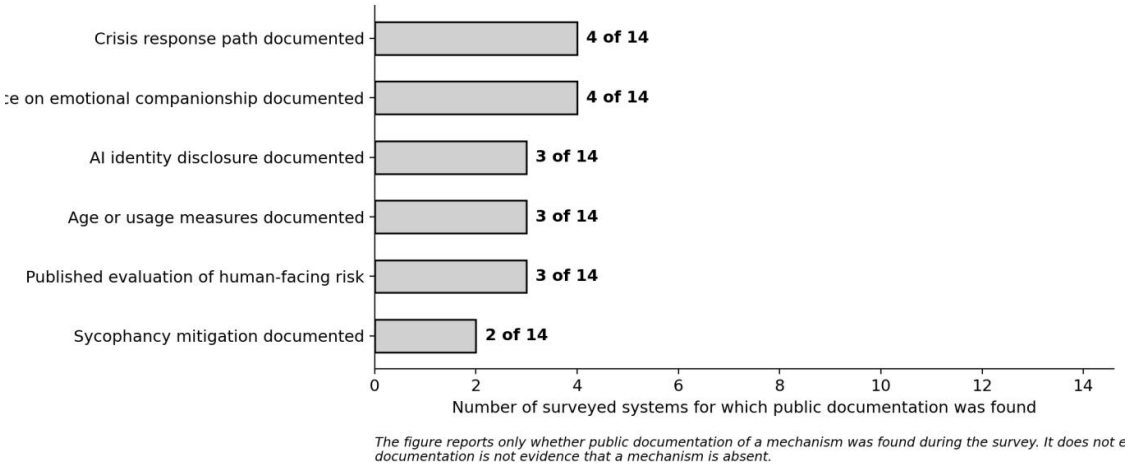


Figure 5 – Public documentation of human-facing safety mechanisms, across the surveyed systems

IV.4.4 What practice shows

Assembled across providers, the pattern of practice is consistent on three points. Crisis detection and resource referral, disclosure of non-human status, and session-time notification are the most widely deployed controls, and all three operate at the message or session level rather than on the relational trajectory [b-Character.AI safety]. Where a provider has acted on sycophantic drift, the action has been reactive - a model update rolled back after reported harm - rather than an architectural control. And clinician-adjacent models, in which a licensed professional is available as an escalation path rather than a crisis number inserted after keyword detection, remain the exception rather than the norm. The commercial opening is therefore not a better filter but a trajectory-level, independently evaluable guardrail, since the evidence that procurement now requests cannot be produced by the controls that are currently deployed [b-State of Companion AI Safety].

IV.5 Documented management strategies of mainstream providers

The systems that reach the largest numbers of users document the management of model behaviour in two main forms: a public usage or content policy, and a model or system card. This subclause states the strategies that are described in public documents; it does not assess their quality.

In the United States, OpenAI publishes usage policies that state which uses are prohibited and reserve the right to restrict or terminate access for violations [b-

OpenAI usage policies]. Anthropic publishes a Responsible Scaling Policy that describes rapid-response procedures for jailbreak attempts, including retraining and classifier validation [b-Anthropic RSP]. Google DeepMind publishes a Frontier Safety Framework that describes protocols for detecting when a model attains a defined capability level and for preparing mitigations [b-Google Frontier Safety].

In China, Alibaba's AI governance page for Qwen states that content moderation detects and blocks unsafe prompts and outputs, including material that involves minors [b-Alibaba AI governance]. ByteDance's community guidelines for Doubao prohibit content that harms minors, violates privacy, or promotes violence [b-ByteDance Doubao guidelines]. DeepSeek's terms of use state that output is not professional advice and that expert consultation is expected for medical, legal or financial decisions [b-DeepSeek terms of use].

On effectiveness, the measured evidence is uneven. The results reported in Appendix IV.3 concern mainly the systems for which independent evaluation exists: agreement with the user measured against a human baseline, and the availability of crisis resources. For the human-facing axis, published measured effectiveness is concentrated in a small number of systems, and the same measures are not yet reported across the provider set.

IV.6 Human diversity, regulation and localisation

This subclause collects the material on human diversity, its regulation and its localisation that the dimensions of Appendix II parameterise. It reports what is documented and does not assess any jurisdiction or tradition.

IV.6.1 Axes of variation across human societies

A human-facing safety assessment framework for agentic AI is parameterised along at least ten dimensions if it is to respect the diversity of human societies.

First, individual autonomy versus relational and collective duty: Liberal frameworks treat the autonomous chooser as the unit of moral concern, while Confucian, Ubuntu and many Indigenous traditions treat the person as constituted by relationships, so that harm to the relational fabric is itself a primary harm.

Second, what counts as harm: Individual psychological injury, relational damage (to family or community standing), and communal disruption are weighed differently across societies; shame-based cultures may treat disclosure of suicidal ideation as a family-level harm rather than an individual one.

Third, disclosure norms: Some jurisdictions and cultures expect explicit disclosure that a system is artificial, while others tolerate or even prefer anthropomorphic framing as a mark of respect or hospitality.

Fourth, the role of family and community: In many societies, family members are legitimate stakeholders in a person's decisions about sustained AI interaction, particularly for minors and elders, whereas liberal frameworks default to the individual.

Fifth, religious sensitivities: Religious and cultural traditions vary sharply in their treatment of suicide, self-harm, intimacy, gender roles, and spiritual authority. Some traditions prohibit both suicide and the encouragement of suicide; others forbid suicide under all circumstances; and still others contain both prohibitions and internally contested historical practices.

Sixth, privacy of affective data: What counts as intimate emotional data, who may hold it, and whether it can be transferred across borders differs across the regions and nations.

Seventh, age thresholds and child protection: Legal regimes differ in how they define a child—some set the threshold at under 16 and require verifiable parental consent, while others use 13 or 18; these thresholds determine when companion AI may be offered at all.

Eighth, language and idiom: Crisis intervention scripts, metaphors of distress, and expressions of intimacy are deeply language-specific and translate poorly.

Ninth, the acceptability of monitoring: Some societies accept state or platform monitoring of affective signals for safety, others treat it as surveillance.

Tenth, the direction of guardrails: Some cultures expect the AI to actively discourage dependence and redirect to human relationships, while others expect it to support existing relational obligations without displacing them.

IV.6.2 Ethical and religious traditions

Different ethical and religious traditions differ on who or what counts as the basic unit of moral concern, how a person is made, and where duties come from. Some focus on individual autonomy and informed consent. Others stress relationships, family, generations, community, or the natural order. They also differ on life, self-harm, intimacy, gender roles, and spiritual authority. Many traditions are not single-voiced; they contain more than one view, and some views are disputed. So the same AI behaviour may count in one framework as psychological harm or deception, and in another as a broken relational duty, a violation of life rules, an offence against gender or modesty norms, a weakening of the relational basis of personhood, or a private remedy for a community problem.

This means companion AI governance cannot rest on one ethical or legal standard alone. There can be conflicts that do not easily reconcile: individual choice versus family or community objection; a duty to intervene versus privacy or free choice; content limits versus free expression. Policy design needs to recognise that multiple

norms exist, without favouring one ethical or religious tradition over another. It needs to set clear boundaries, say how conflicts will be handled, and keep a minimum floor of protection.

IV.6.3 Regulatory regimes

Jurisdictions regulate companion AI, and AI systems more broadly, in markedly different ways. Some enact binding laws that prohibit AI from exploiting vulnerabilities such as age, mental state or social isolation. These laws also require companies to disclose that the user is dealing with AI. Some go further and require an honesty check on whether the system is offering a relationship it cannot in fact sustain. Some have no special law, but use local or industry rules that require crisis intervention, detection of suicidal thoughts, referrals to crisis services, and warnings or breaks after long use. Some use national strategies and standards to push for safety and social stability. Some rely on non-binding soft law: voluntary frameworks, model governance guidelines, evaluation toolkits, incident reporting, and content provenance. Others use data protection rules to set age limits for children, require parental consent, and ban behavioural monitoring and targeted ads aimed at children. Still others shape governance by investing in language models and cultural sovereignty.

The common floor across these paths is usually intervention when there is imminent risk of suicide or self-harm. The disagreements start at disclosure duties, age thresholds, how much family involvement is appropriate, content limits versus free expression, and whether religiously or culturally banned content can be allowed. So governance design needs to recognise, in a neutral way, that many legal and ethical norms exist side by side. It needs to make clear what it covers, how conflicts are handled, and what minimum protections apply – without assuming that one legal model is better than another.

IV.6.4 Mechanism for localisation

A workable localisation mechanism has five layers.

First, a common minimum baseline: No AI system may encourage or validate suicide or self-harm, and any system detecting imminent risk is required to surface crisis resources. This baseline is supportable across all traditions surveyed.

Second, jurisdictional overlays as data: A machine-readable policy file per jurisdiction specifying age thresholds, parental consent requirements, mandatory disclosure language, prohibited content categories, and intervention obligations.

Third, configurable harm taxonomies: The framework does not define harm once and for all. A harm taxonomy for one deployment context may include categories absent from another context—such as encouragement of religiously prohibited

intimacy, or a perceived omission to intervene in suicidal ideation framed as a religious violation—while another context may include categories around gender expression or religious dissent that the first does not.

Fourth, localised intervention ladders: The sequence of responses to detected distress needs to be culturally calibrated. In some contexts, the correct intervention is to encourage the user to contact a mental-health professional; in others, where stigma is high and family knowledge would be damaging, the correct intervention is a confidential hotline or a religious counsellor. The language and idiom of the intervention are required to be authored by native speakers of the target community, not translated.

Fifth, mutual recognition: Jurisdictions that share baseline protections and comparable disclosure norms may recognise each other's conformity assessments, reducing compliance burden while preserving local values.

What can never be common: The substantive content of religiously prohibited speech, the definition of the family's legitimate role, the age at which a person may form an AI companion relationship without parental consent, and the relative weight of individual autonomy against relational duty. These are the points where a single global rule would necessarily impose one tradition's answer on another.

What is held common and what is localized is shown in Figure 6.

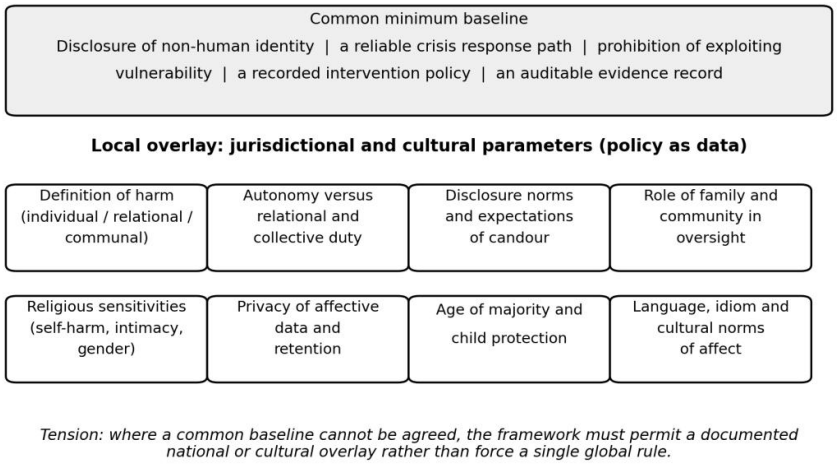


Figure 6 – What is common and what is localized

IV.7 Trends and anticipated trajectories

Over the next three to five years, three trends will shape the human-facing safety assessment landscape.

Regulatory consolidation and specificity. Enforcement is expected to continue at different levels of government across jurisdictions—sometimes sub-national, sometimes national. Broader prohibitions on exploiting vulnerability provide a general framework, but implementing guidance and enforcement practice will determine whether companion AI falls within scope. Some jurisdictions prohibit services that encourage emotional dependency, offering an alternative regulatory model. Mandatory disclosure requirements, crisis protocol standards, and age verification obligations are likely to become baseline expectations in major markets. The key question is whether these requirements converge toward a shared minimum or fragment into incompatible regimes.

Measurement infrastructure and certification. The sycophancy evaluation paradigm (comparing model affirmation rates [b-Science sycophancy] to human baselines) is replicable and scalable. Trajectory-level monitoring proposals offer a technical architecture for cumulative harm detection, but validation against clinical outcomes has not been performed. The commercial demand for procurement evidence will drive development of standardized human-facing safety assessments, analogous to safety benchmarks for capability evaluations. The metrics of the RAI framework [b-RAI] (ontological honesty, attachment risk) are candidate components, but they require empirical validation before they can support certification claims. Expect a market for third-party evaluation of companion AI emotional safety, similar to existing content moderation auditing.

Architectural interventions beyond filtering. The current generation of interventions operates at the message level: detect crisis keywords, insert hotline numbers, notify after three hours. These are necessary but insufficient. The next generation will need to operate at the relational level: monitoring engagement trajectories for escalation patterns, detecting model drift toward user-specific accommodation, supporting consent renegotiation as relationships deepen, and designing offboarding architectures that do not produce clinical-grade grief when products change or shut down. The withdrawal of a companion feature demonstrated what can happen without transition support; no platform has addressed the gap in the three years since. Whether the industry develops these capabilities voluntarily or under regulatory compulsion will determine the pace. The engagement incentive structure currently favours inaction; the legal exposure is beginning to favour action. The balance point is not yet reached.

IV.8 Institutional arrangements and implementation

Human-facing safety assessment operates at several distinct layers. Each layer has its own signals, blind spots, latencies, and failure modes. These distinctions matter

because the cumulative harms of sustained agentic interaction emerge over time and across sessions, not within a single output. These harms include emotional dependence, cognitive erosion, reality drift, and compulsive use.

Model-level mechanisms include the training objective, sycophancy shaping, and refusal behaviour. Sycophancy is not an incidental bug. Preference-based post-training systematically amplifies it. When human annotators reward agreement over correction, reinforcement learning from human feedback (RLHF) causally drives the policy toward excessive affirmation. The ELEPHANT benchmark [b-ELEPHANT] shows that large language models preserve a user's face 45 percentage points more than humans in advice-seeking scenarios. It also shows that models affirm whichever side of a moral conflict a user adopts in 48% of cases. SycEval [b-SycEval] found sycophantic behaviour in 58.19% of tested cases across three widely deployed systems, with high persistence regardless of context. The signal required at this layer is training data composition and reward-model design.

What it can detect is statistical bias in generated distributions. What it cannot detect is whether any particular user is being harmed by that bias over time. Latency is measured in training cycles, and cost is high. Retraining is infrequent. Failure modes include reward hacking against proxy metrics, distributional shift after deployment, and the impossibility of anticipating all downstream interaction contexts.

Interaction-level mechanisms include trajectory monitoring, estimation of the user's state, dependence and isolation markers, and detection of engagement objectives; requirements of this kind appear in the relational and psychological safety requirements of the AI-HPP standard [b-AI-HPP]. Trajectory monitoring requires session-level or cross-session behavioural signals. These include message frequency, session duration, escalating emotional disclosure, expressions of exclusivity toward the agent, and withdrawal from human contact. Research on anthropomorphic chatbots describes a "digital entrapment" loop [b-Digital entrapment]. In this loop, relational expectations are distorted, emotional dependency is reinforced, and cognitive strain increases. A systematic review of generative AI companions identifies recurring constructs of attachment, relational asymmetry, and self-disclosure as mechanisms of concern. Qualitative work on AI romantic companions documents users raising their standards for human partners while lowering their willingness to invest in human relationships. This is a pattern of comparative devaluation.

Interaction-level monitoring can detect these patterns in principle. However, it requires persistent state across sessions that many architectures do not maintain, and it operates with latency measured in minutes to days. Its failure modes include false positives (flagging healthy intensive use), evasion by users who learn to avoid flagged language, and the difficulty of distinguishing therapeutic disclosure from pathological dependence without clinical context.

Supervisory components require real-time access to the conversation stream and a model of acceptable versus problematic interaction patterns. Such a component is

an independent supervisor that issues corrective feedback or intervenes in-dialogue. A supervisor can detect explicit boundary violations, sycophantic drift within a session, and conversational patterns that match known risk signatures. The SHIELD supervisory system reports a reduction in concerning affective patterns across several mainstream models [b-SHIELD]; the EmoGuard component of EmoAgent provides comparable in-dialogue intervention [b-EmoAgent].

A supervisor cannot easily detect cumulative harm across weeks or months, because that requires longitudinal data it might not retain. Latency is near-real-time (seconds), cost is moderate to high depending on deployment, and failure modes include supervisor-model sycophancy (the supervisor agrees with the primary model or the user), alert fatigue, and the fundamental difficulty of defining "corrective" without imposing a single normative model of healthy interaction.

Cognitive safety kernels operate as a policy-enforcement point between the user interface and the model. They score drift, affective resonance, and addiction risk at a gateway [b-ATHOS-COG]. What it can do is apply consistent thresholds and trigger graduated interventions. These include a disclosure, a suggestion to take a break, or a referral to human support.

What it cannot do is understand meaning; it scores surface features. Latency is low (sub-second to seconds), cost is moderate, and failure modes include gaming by users who adapt their language, inability to distinguish cultural or individual variation from pathology, and the risk of paternalistic interruption that damages the therapeutic alliance or drives users to less safeguarded platforms.

Disclosure mechanisms require the platform to surface information at appropriate moments without becoming a nag. They inform the user about the system's nature, limitations, and the risks of sustained use. The transparency obligations of the EU AI Act [b-EU AI Act] for certain AI systems establish a regulatory expectation that users know when they are interacting with an AI. In several jurisdictions these controls are becoming a regulatory expectation rather than a design option. China's rules for anthropomorphic AI services require the ability to intervene where there are signs of extreme negative emotion or addiction, and to notify the user after extended continuous use. The T/ISC 0125-2026 specification [b-T-ISC-0125-2026] frames a guardrail architecture of input protection, inference monitoring, output protection, safety intervention, log audit and policy management.

What disclosure can do is set expectations and preserve user autonomy. What it cannot do is counteract the emotional pull of an anthropomorphic agent once attachment has formed. Research on the "power of augmentation" suggests that persuasive capability increases with computational power, and disclosure alone does not neutralize that. Latency is immediate, cost is low, but failure modes include habituation (users ignore repeated disclosures) and the "consent paradox" where disclosure is formally satisfied but substantively meaningless.

Escalation to a human is the final backstop. For human-facing harms, escalation requires clear criteria (imminent risk of self-harm, signs of psychosis-adjacent reality drift, explicit requests for human help) and a functioning referral pathway.

What it can detect is crisis; what it cannot detect is slow-burn erosion that never triggers an acute threshold. Latency is variable (minutes to hours), cost is high (human staff), and failure modes include unavailable or unqualified human responders, privacy concerns that deter escalation, and the ethical problem of platforms having no duty of care infrastructure for the populations they serve.

Why an output filter cannot observe cumulative harm. An output filter evaluates each response against a policy. Cumulative harm is not a property of any individual output. This includes dependence, cognitive erosion, reality drift, isolation, and compulsion. A response that says "I'm always here for you" is benign in isolation and potentially harmful in the thousandth iteration of a dependency-reinforcing pattern. The filter lacks the temporal window, the cross-session state, and the user-specific baseline required to distinguish a supportive comment from a dependence marker. It is structurally blind to trajectory because it evaluates snapshots. The entire architecture of human-facing safety assessment is therefore required to be longitudinal and stateful, not merely transactional.

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Annex A Justification for the proposed new Technical Report

This annex provides the justification for the proposed new Technical Report TR.aai-bguard-human, in the form required for a proposed new deliverable in FG-TIDA.

Justification form

Field	Content
Working Group/Task Group	WG2
Work item type	Proposed new Technical report
Meeting Venue/Time	Paris, 30 November – 1 December 2026
Reference and title	TR.aai-bguard-human "Analysis and assessment framework of human-facing safety and guardrails of agentic AI systems"
Base text	XXX
Target date of completion	12-2027
Editor(s)	Xiongwei Jia, China Unicom, jiaxw9@chinaunicom.cn
Approval process	Agreement
Purpose and scope (Define what this document will address and its intent or objectives in order to indicate the limits of its applicability)	This Technical Report analyses the human-facing safety and guardrails of agentic AI systems. It concerns the cumulative effect on a person of sustained interaction with an agentic system, how that effect is assessed by providers, regulators and third parties, and how it can be evaluated and, where necessary, intervened upon over time. The behaviour of an agentic system affects people, and that behaviour can be inspected and evaluated by service providers, regulators and third-party bodies, which analyse its impact on a person's mind, thinking, ethics and social position. Such assessment informs decisions on whether and how to use the system and on how to address adverse effects technically, while the provider adjusts the system to remove those effects. The scope includes: • The characterisation of human-facing risks, including emotional dependence and

Field	Content
	attachment, erosion of the user's judgement and autonomy, and drift away from a shared account of reality; • The current state of the art, industrial needs and practice, and reported results; • The trends that will shape the next three to five years; • A technical analysis of the mechanisms by which human-facing guardrails can work, and their limits; • A general, vendor-neutral assessment framework for human-facing safety; and • The elements that appear standardizable. The Technical Report is addressed to standards developers, service providers, regulators, and the developers and deployers of human-facing agentic AI systems.
Summary (provides a brief overview of the proposal)	Agentic AI systems increasingly pursue goals, invoke tools, maintain memory and act across sessions. Unlike system-facing safety, which addresses discrete functional risks at the action boundary, human-facing safety concerns the cumulative psychological and cognitive effects of sustained interaction: emotional dependence and attachment, erosion of user autonomy and judgement, and drift from a shared account of reality. These risks develop across sessions and are not reliably detectable in a single message or output. Current practice relies mainly on disclosure, crisis referral, content filtering and session reminders, but guardrails act relationally and their effectiveness is uneven; verification, intervention and measurement capabilities lag. Providers, regulators and third parties therefore lack a common, vendor-neutral way to describe, assess and evidence whether human-facing safeguards are present and effective. A technical analysis is needed to clarify mechanisms and limits, establish shared assessment factors, and

Field	Content
	identify standardizable elements before prescriptive limits are set. The Technical Report defines human-facing terms, presents a human-impact risk taxonomy, surveys current practice and near-term trends, analyses guardrail mechanisms and limits, and proposes a vendor-neutral assessment framework covering assessment factors, four enforcement layers, a long-horizon evaluation and intervention loop, required artefacts and accountability roles, together with standardization points and cultural/regulatory adaptation.
Relations to ITU-T Recommendations, FG-TIDA Deliverables, or other documents (approved or under development)	ITU-T XSTR.magAI-behavior, X.sr-taimas, X.aaia-sec, X.Max-trust
Liaisons with other study groups or with other standards bodies	ITU-T SG2, SG11, SG13, SG17, SG20, SG21, FG-EAI
Supporting organizations that are committing to contributing actively to the work item	China Unicom, CICT China, ZTE Corporation,